Escuela Técnica Superior de
INGENIERÍA DE SEVILLA

Estudio de estabilidad global de flujos laminares alrededor de perfiles NACA 0012 y 2412

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Índice

1.- Introducción

2.- Formulación del problema
- Ecuaciones y condiciones de contorno
- Parámetros de control. Número de Reynolds y ángulo de ataque
- Soluciones básicas 2D estacionarias
- Perturbación lineal. Modos estable e inestable
- Método numérico
- Análisis de convergencia de flujo
- Análisis de convergencia de autovalores
-Elección/Optimización de la malla

3.- Resultados
- NACA 0012
- NACA 2412

4.- Comparación con ANSYS FLUENT
- Definición del problema
- Comparación de resultados
- Régimen transitorio

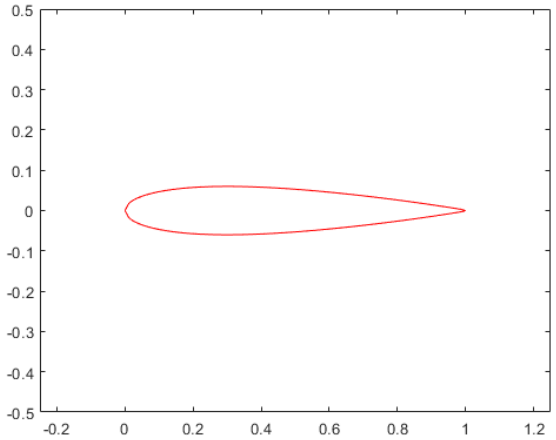
5.- Comparación NACA 0012 vs NACA 2412
- Flujo básico
- Estabilidad

6.- Conclusiones

7.- Bibliografía

1.- Introducción

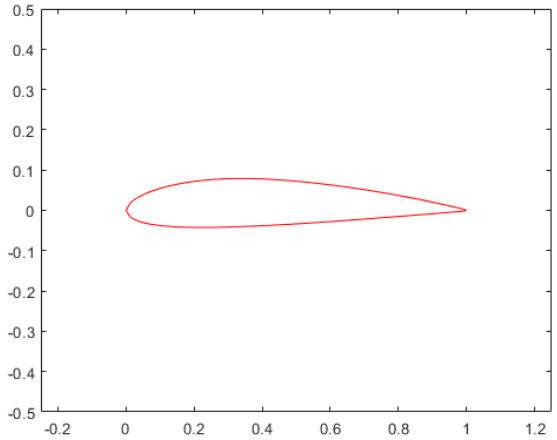
NACA 0012



- Estabilizador horizontal de aeronaves
- Hélices
- Aeronaves no tripuladas (UAV)
- Experimentos y pruebas
- Ejemplo de aeronave

Bede BD-8

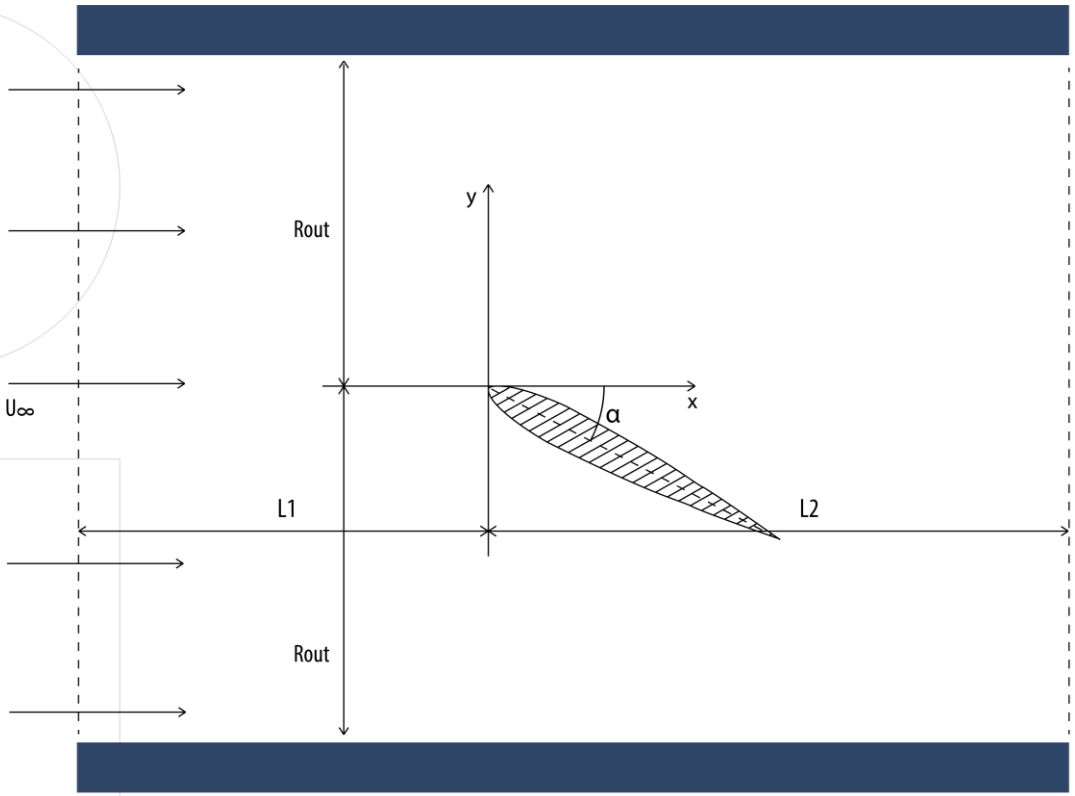
NACA 2412



- Aviones ligeros y de entrenamiento
- Planeadores
- Drones y vehículos aéreos no tripulados
- Experimentos aerodinámicos
- Ejemplo de aeronave

Cessna 152

2.- Formulación del problema. Ecuaciones



Ecuaciones de Navier-Stokes:

$$\frac{\partial v_x}{\partial x}+\frac{\partial v_y}{\partial y}+\frac{\partial v_z}{\partial z}=0$$

$$\frac{\partial v_x}{\partial t}+v_x\frac{\partial v_x}{\partial x}+v_y\frac{\partial v_x}{\partial y}+v_z\frac{\partial v_x}{\partial z}=-\frac{\partial p}{\partial x}+\frac{1}{Re}\left(\frac{\partial^2 v_x}{\partial x^2}+\frac{\partial^2 v_x}{\partial y^2}+\frac{\partial^2 v_x}{\partial z^2}\right)$$

$$\frac{\partial v_y}{\partial t}+v_x\frac{\partial v_y}{\partial x}+v_y\frac{\partial v_y}{\partial y}+v_z\frac{\partial v_y}{\partial z}=-\frac{\partial p}{\partial y}+\frac{1}{Re}\left(\frac{\partial^2 v_y}{\partial x^2}+\frac{\partial^2 v_y}{\partial y^2}+\frac{\partial^2 v_y}{\partial z^2}\right)$$

$$\frac{\partial v_z}{\partial t}+v_x\frac{\partial v_z}{\partial x}+v_y\frac{\partial v_z}{\partial y}+v_z\frac{\partial v_z}{\partial z}=-\frac{\partial p}{\partial z}+\frac{1}{Re}\left(\frac{\partial^2 v_z}{\partial x^2}+\frac{\partial^2 v_z}{\partial y^2}+\frac{\partial^2 v_z}{\partial z^2}\right)$$

2.- Formulación del problema. Solución

$$\mathbf{V}(\mathbf{x},\mathbf{y}) = \mathbf{V}_b(x,y) + \varepsilon \mathbf{V}_p(x,y)e^{-i\omega t} + \text{c.c.}$$

- Solución básica estacionaria:

$$\mathbf{V}(\mathbf{x},\mathbf{y}) = \mathbf{V}_b(x,y)$$

"Mainsolid2tha"

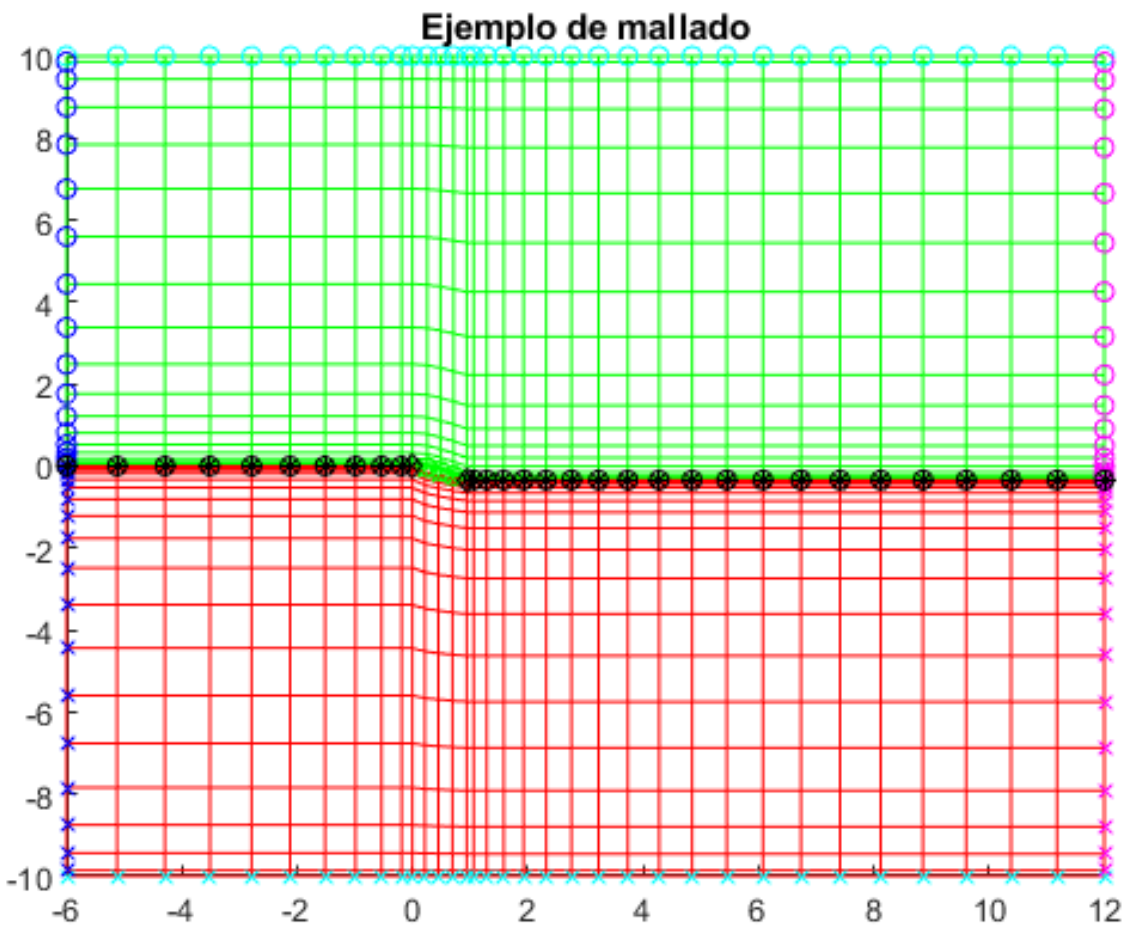
- Perturbación lineal:

$$\omega = \omega_r + i\omega_i$$

"Main_eigen3Dsolid"

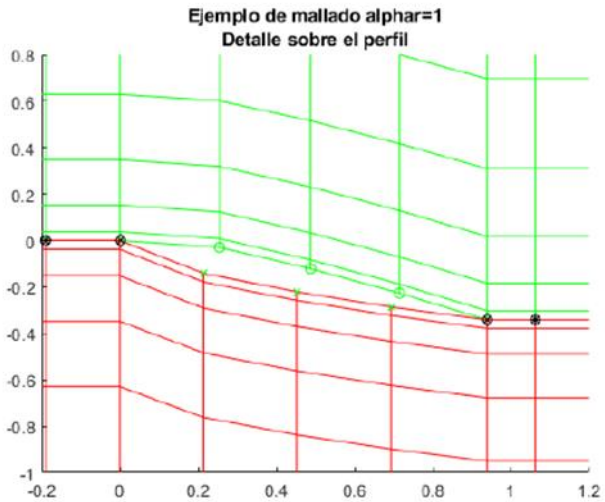
  - Modo estable: $\omega_i < 0$
  - Modo inestable: $\omega_i > 0$

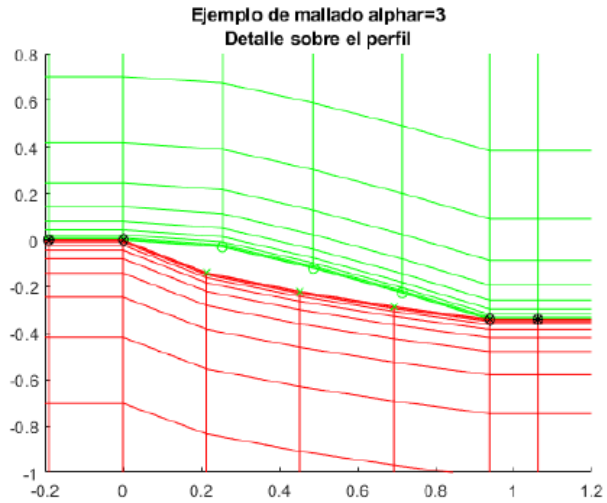
2.- Formulación del problema. Mallado



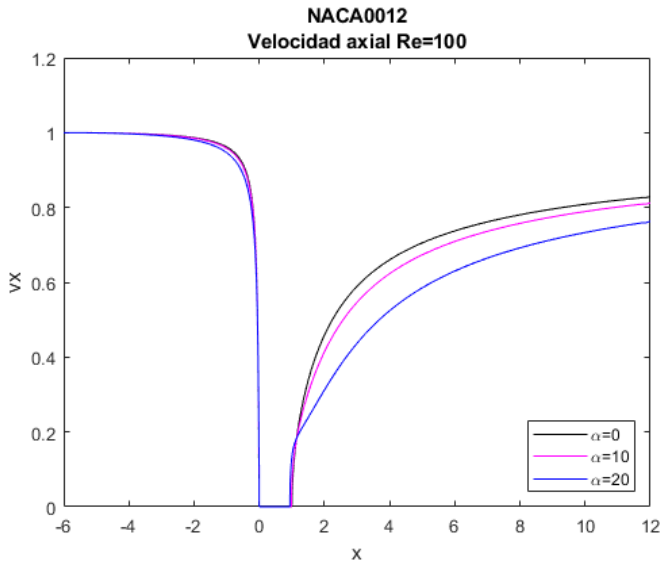
Parámetros:

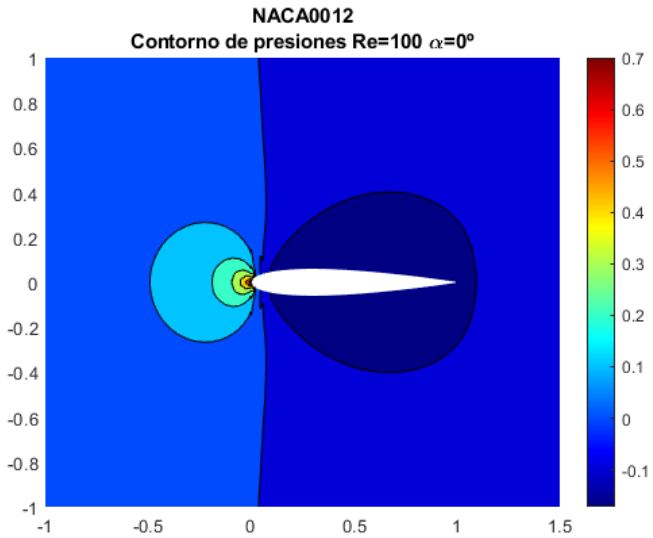
- L1 y L2
- Rout
- nz1, nz2 y nz3
- nrA y nrB
- alphar

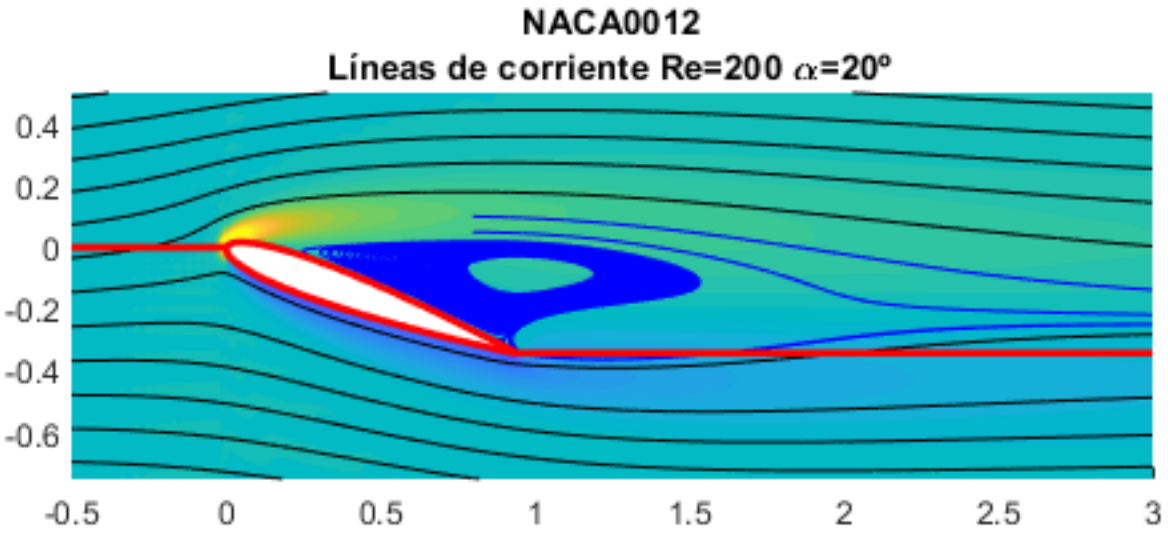




3.- Resultados. NACA 0012. Flujo básico







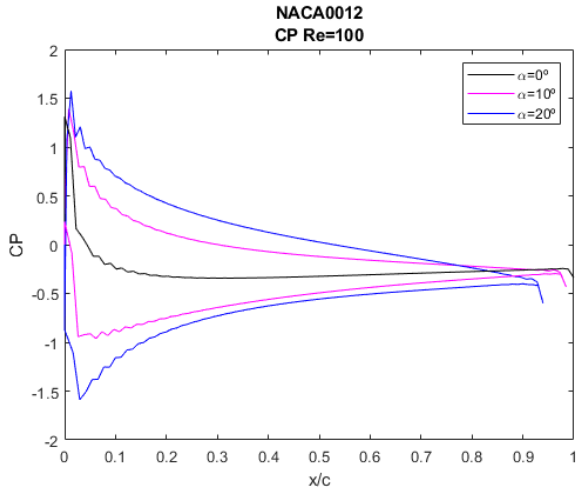
3.- Resultados. NACA 0012. Flujo básico

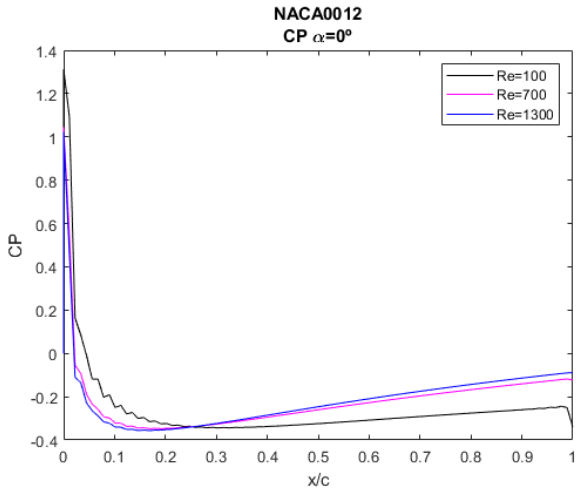
Coeficientes

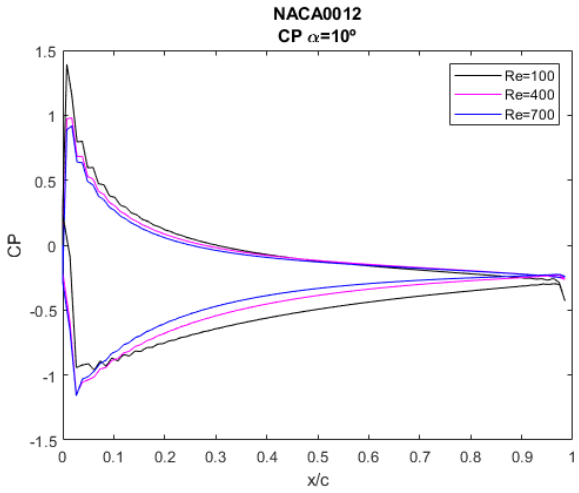
- Coeficiente de presión (C_p) : $C_p = \frac{P - P_\infty}{\frac{1}{2} \cdot \rho \cdot U_\infty^2}$

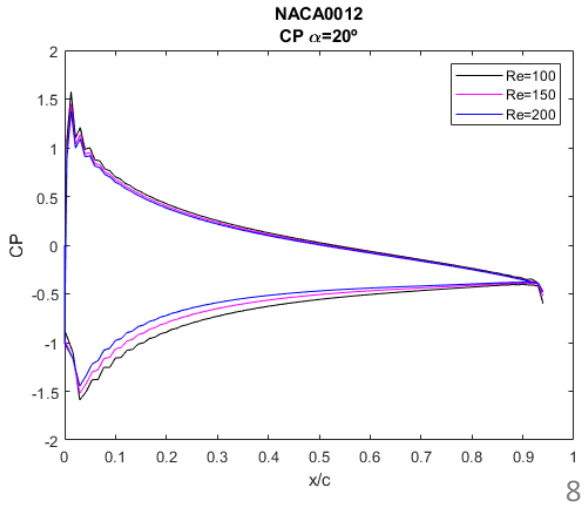
- Coeficiente de sustentación (C_l) : $C_l = \frac{l}{\frac{1}{2} \cdot \rho \cdot U_\infty^2 \cdot c}$, $l = l_P + L_v$

- Coeficiente de resistencia (C_d) : $C_d = \frac{d}{\frac{1}{2} \cdot \rho \cdot U_\infty^2 \cdot c}$, $d = d_p + d_v$

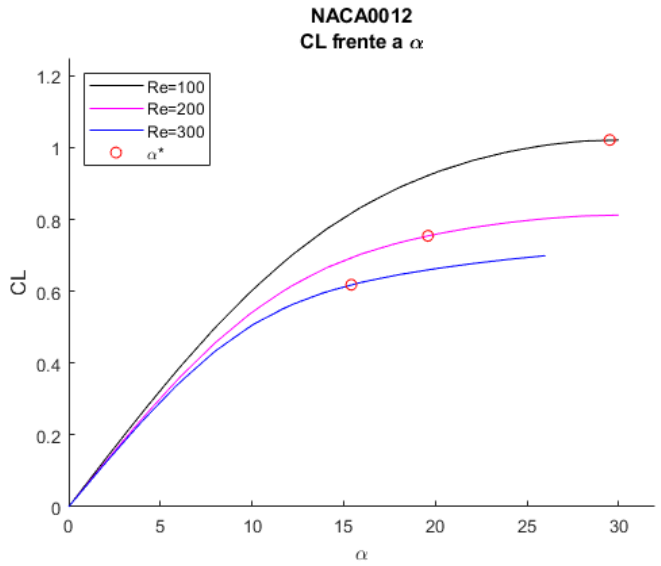


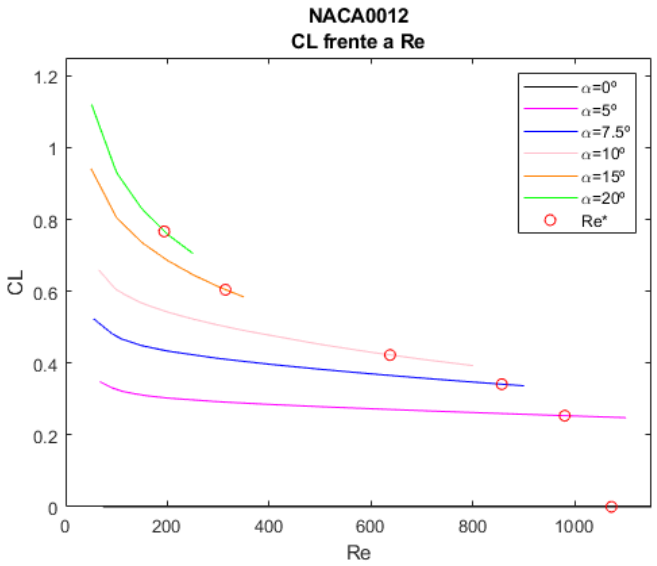


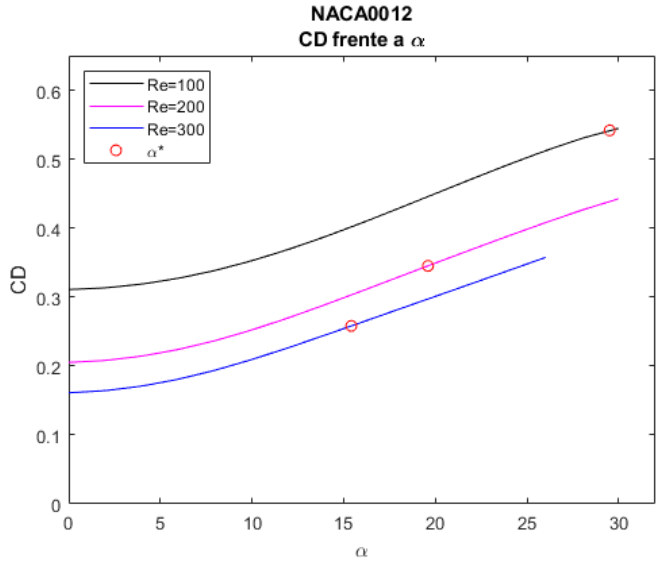


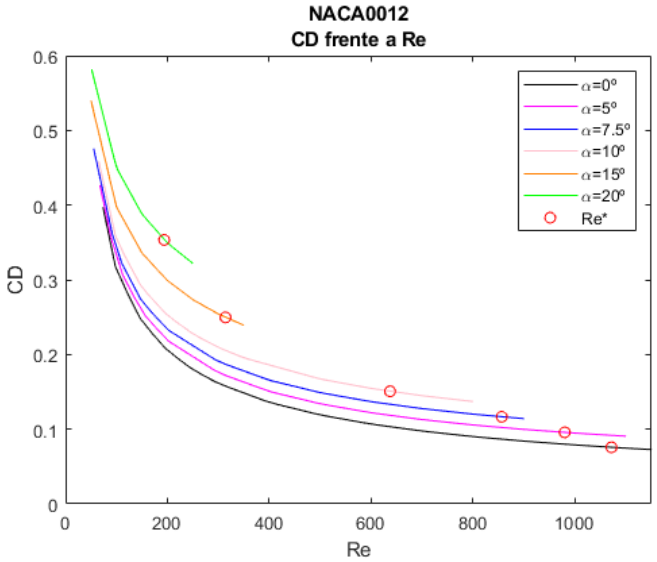


3.- Resultados. NACA 0012. Flujo básico









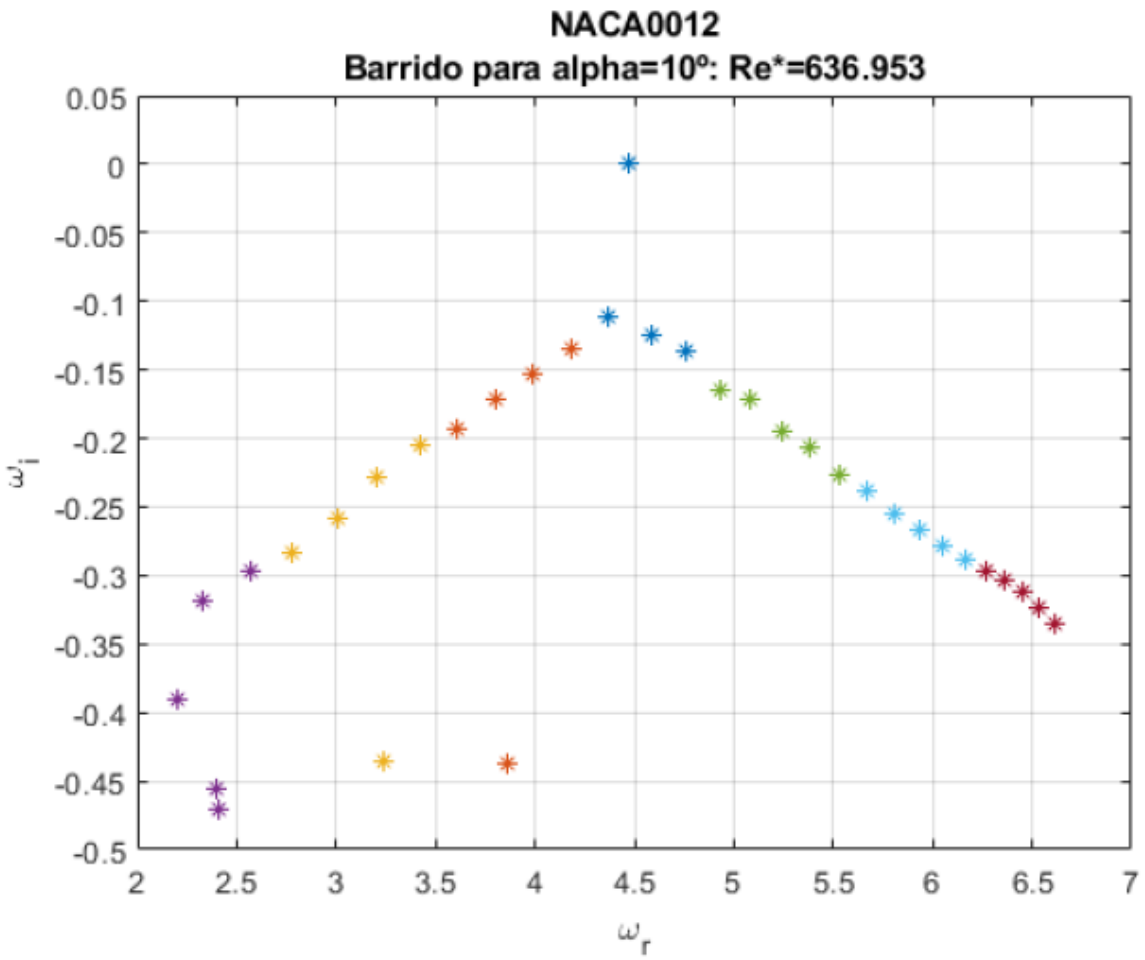
3.- Resultados. NACA 0012. Estabilidad

Número de Reynolds crítico

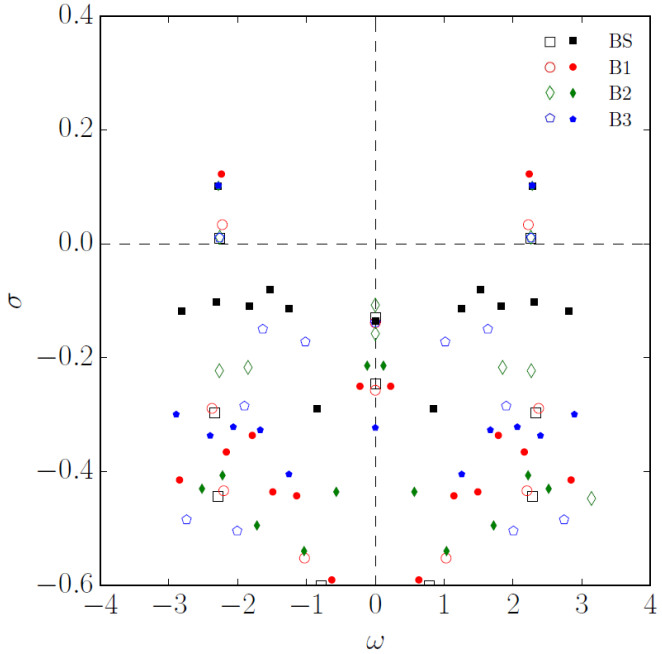
α	Re^*
0º	1071.500
5º	979.750
7.5º	856.211
10º	636.953
15º	313.895
20º	193.652

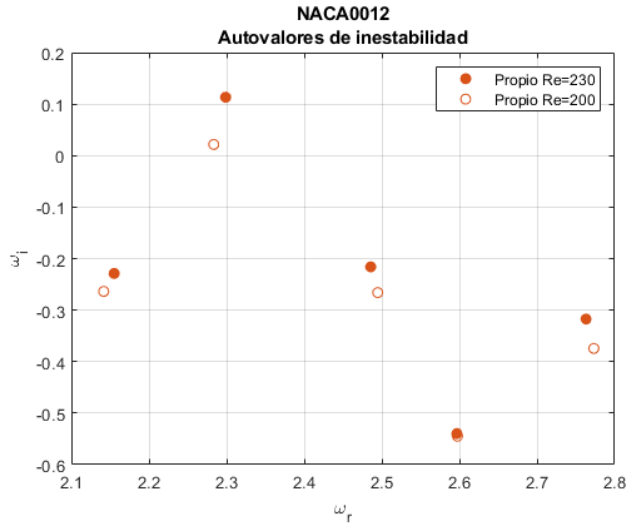
Ángulo de ataque crítico

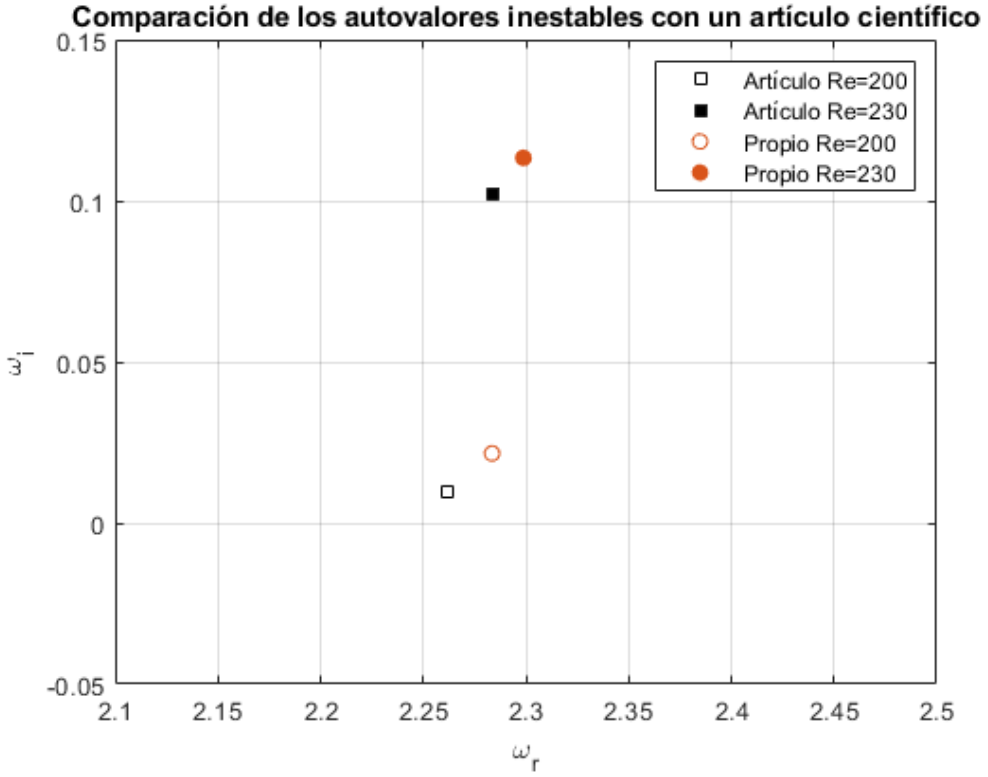
Re	α^*
100	29.509º
200	19.598º
300	15.415º



3.- Resultados. NACA 0012. Artículo.

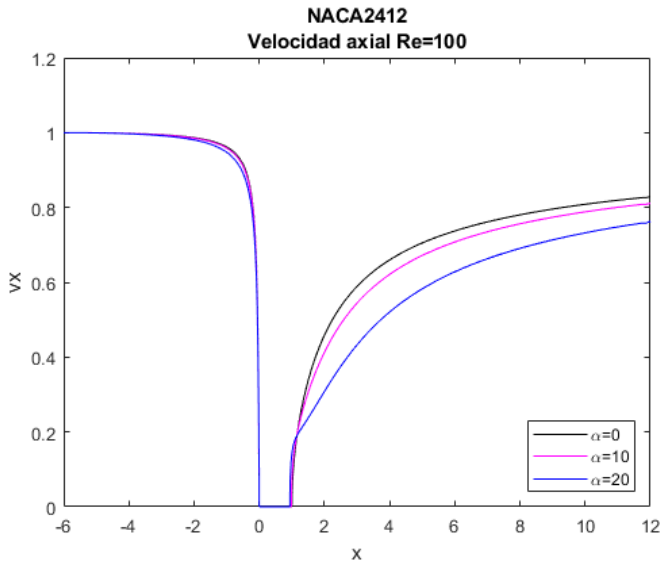


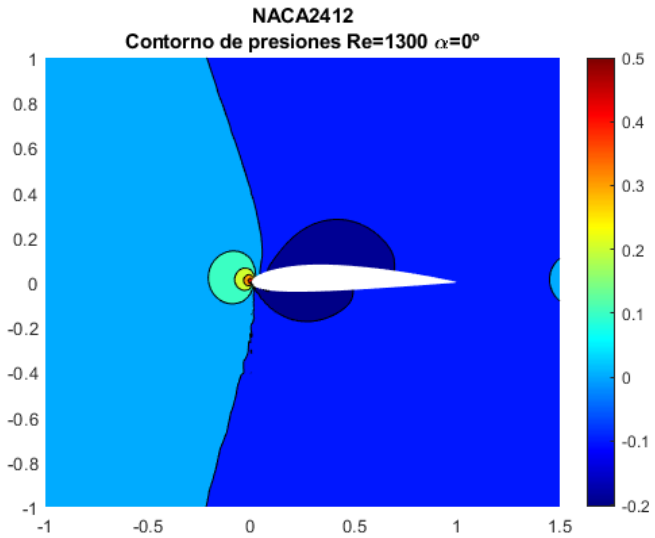


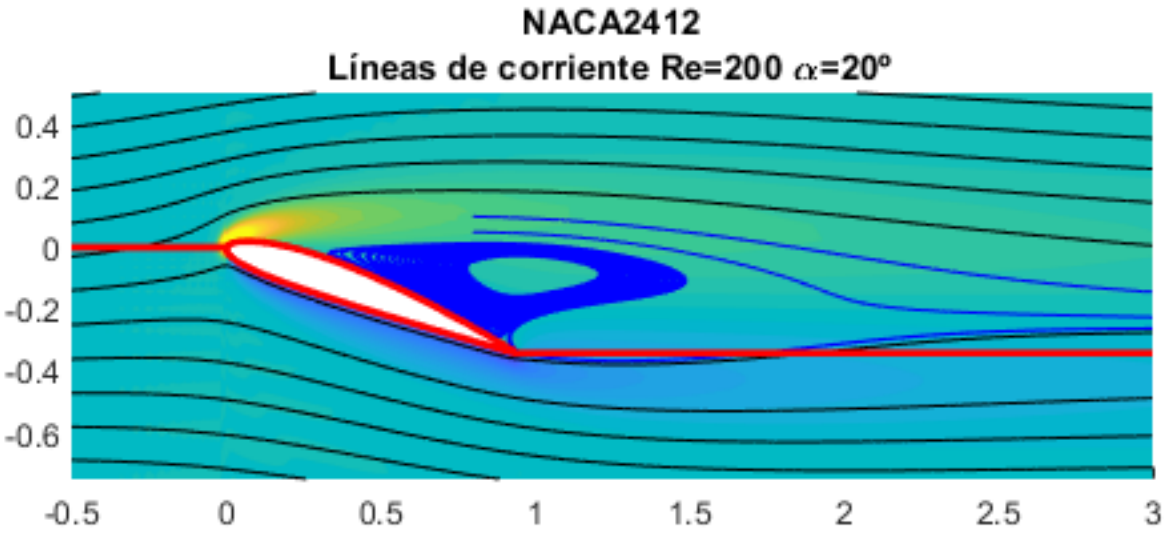


Error relativo	$Re = 200$	$Re = 230$
ε_{ω_r}	0.0098	0.0064
ε_{ω_i}	1.1851	0.1120

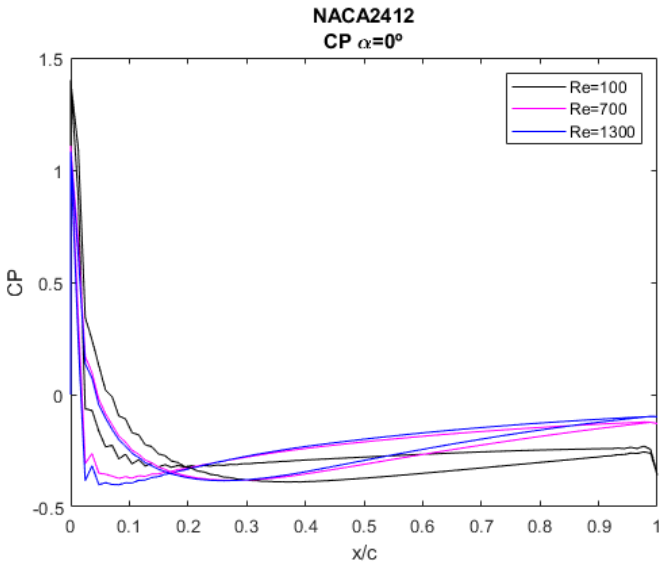
3.- Resultados. NACA 2412. Flujo básico

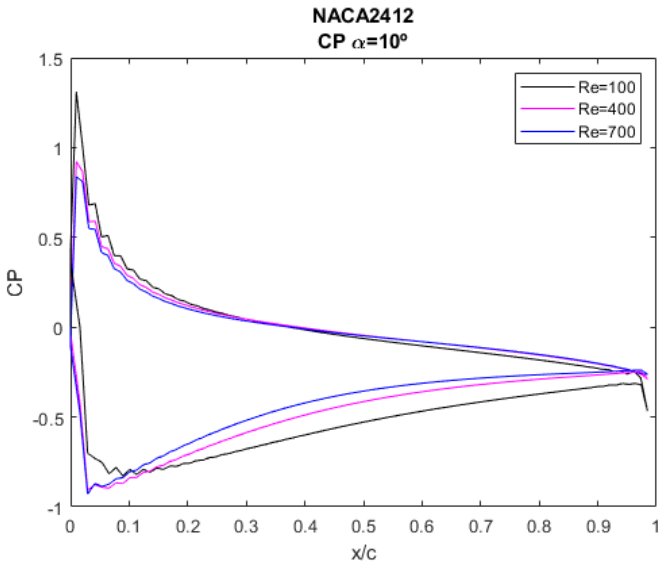


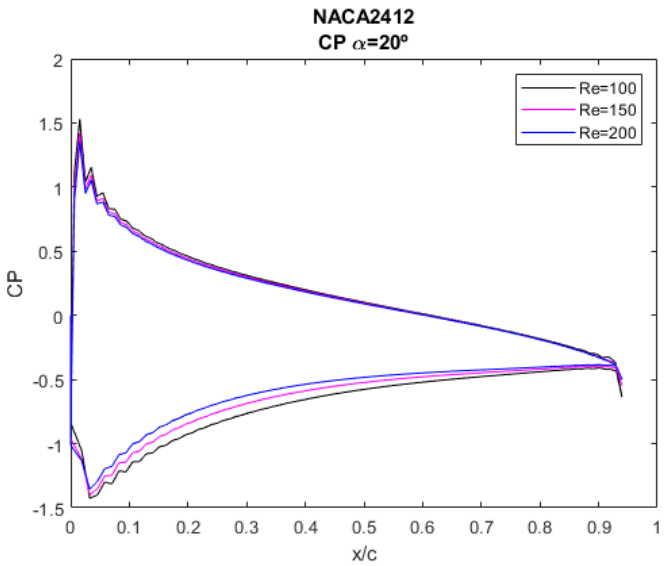


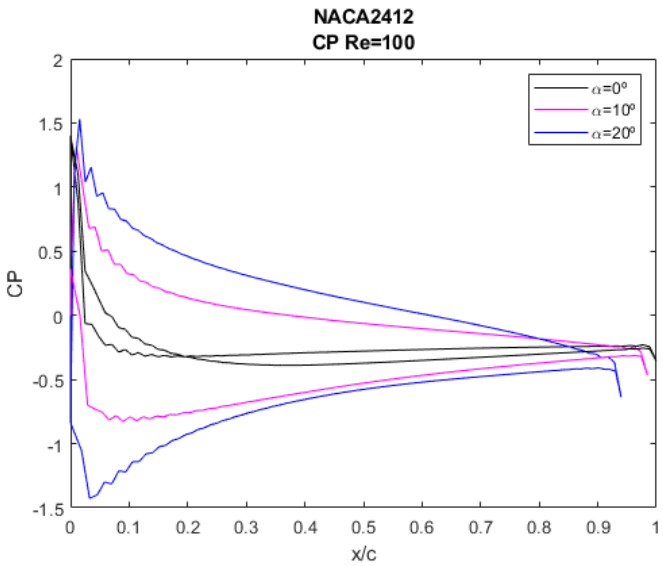


3.- Resultados. NACA 2412. Flujo básico

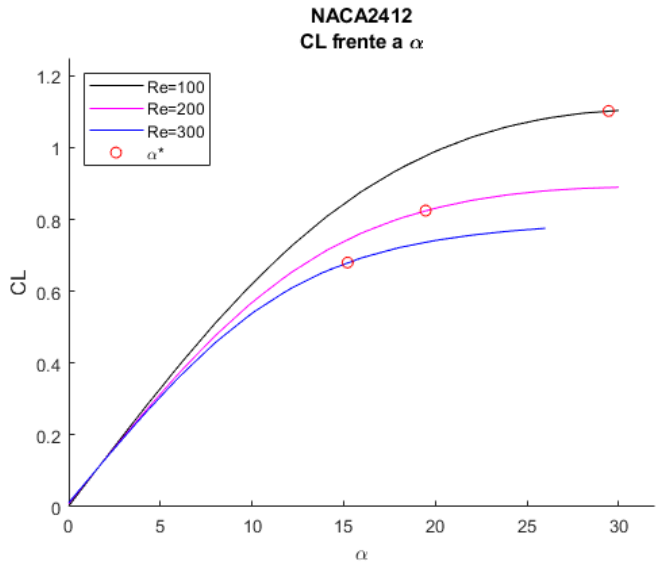


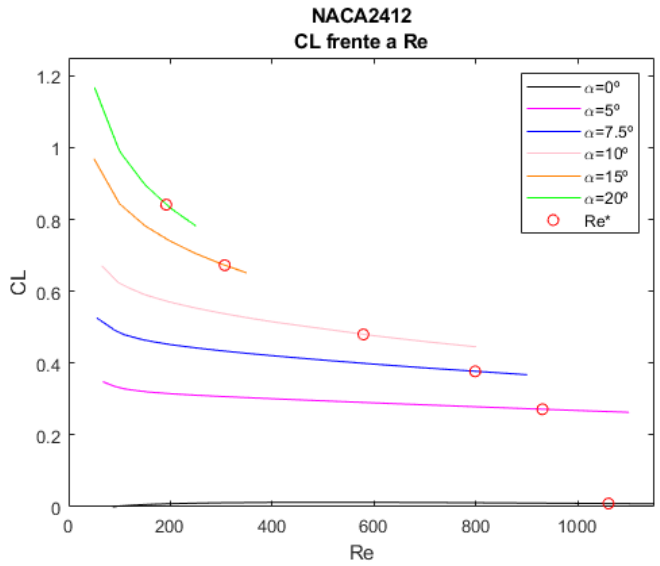


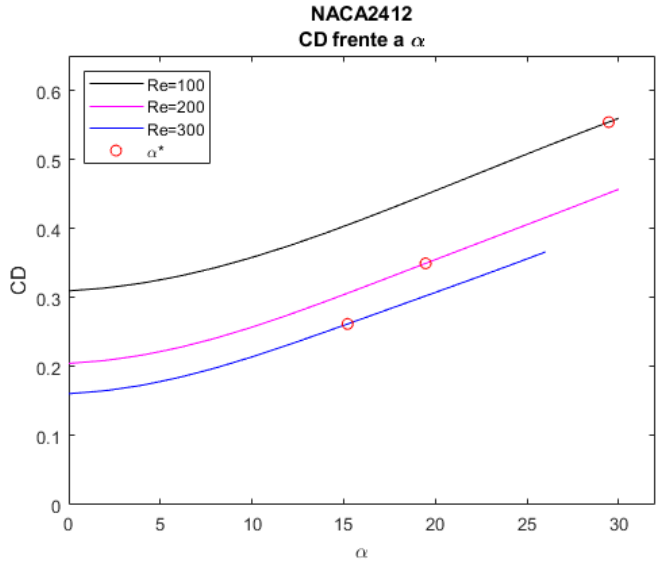


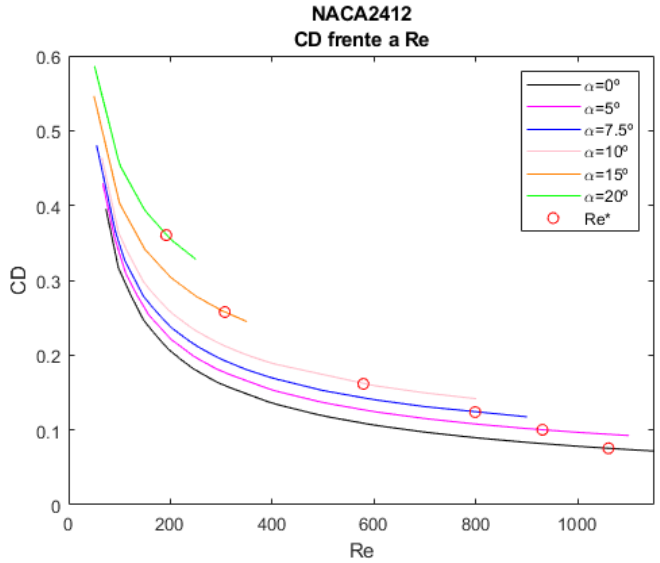


3.- Resultados. NACA 2412. Flujo básico









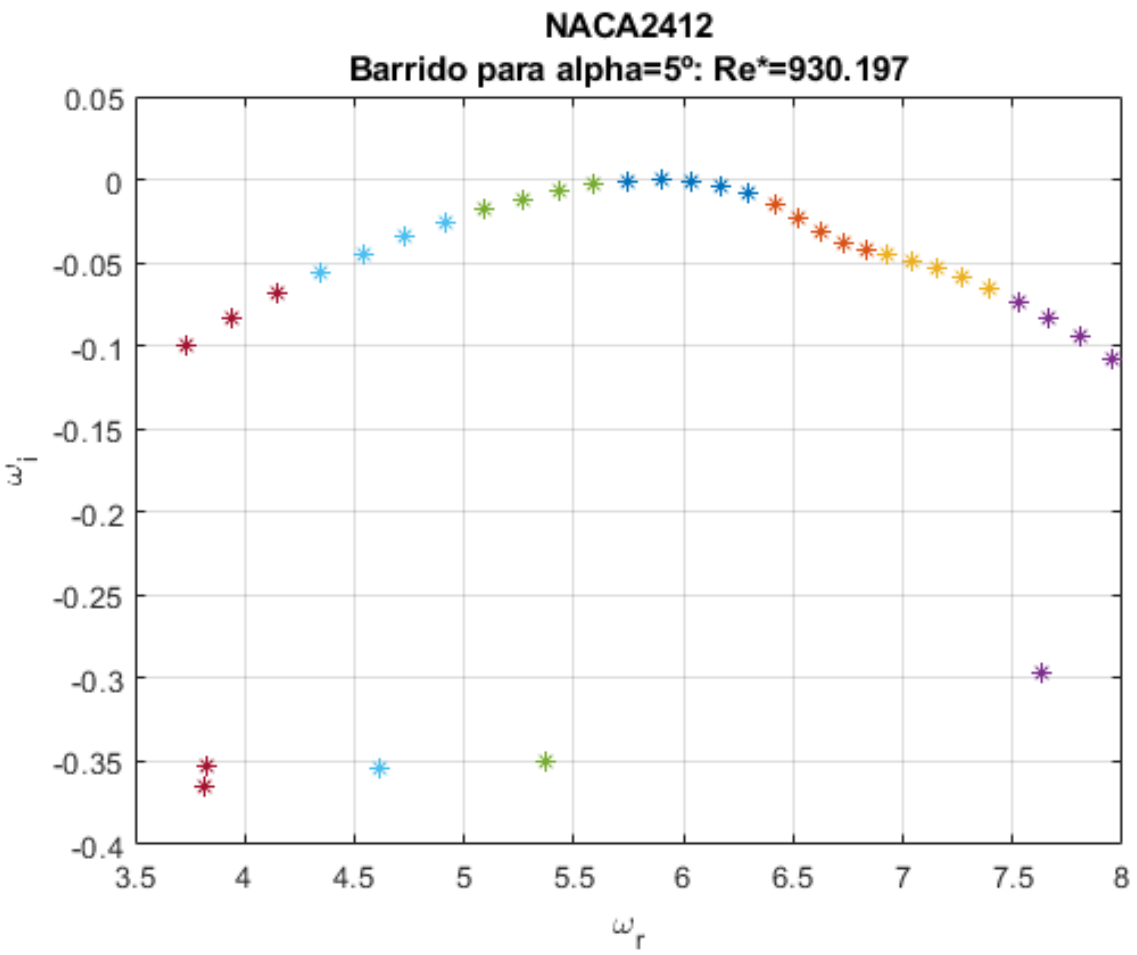
3.- Resultados. NACA 2412. Estabilidad

Número de Reynolds crítico

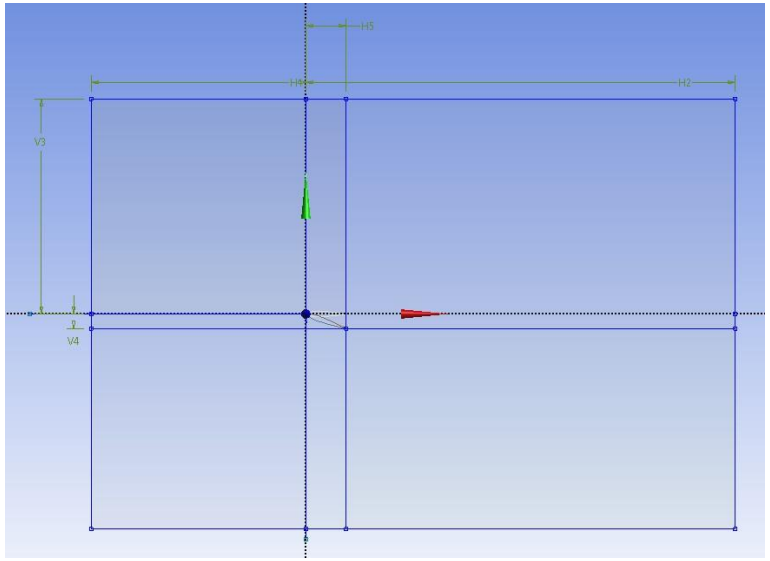
α	Re^*
0º	1059.780
5º	930.197
7.5º	798.090
10º	578.706
15º	306.594
20º	191.680

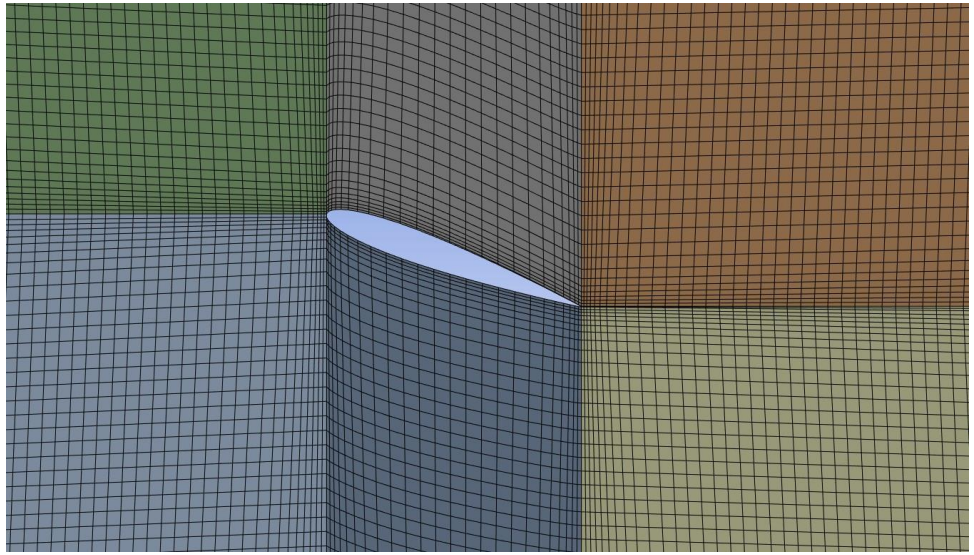
Ángulo de ataque crítico

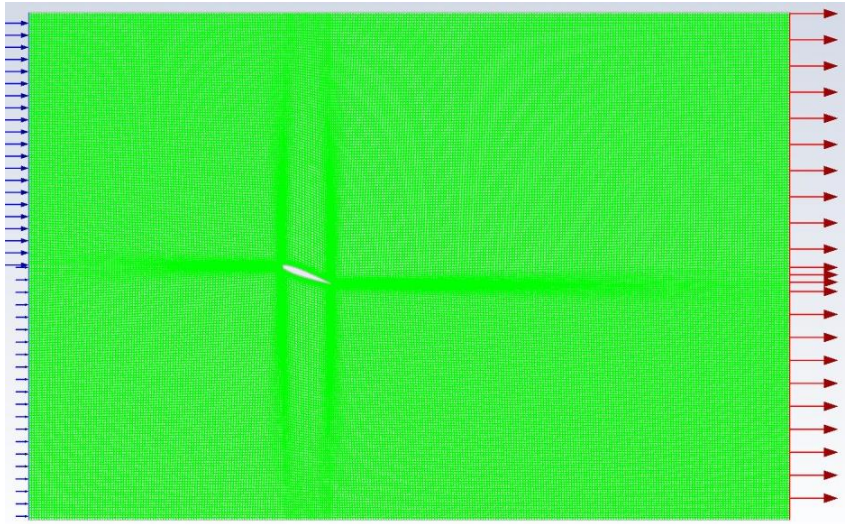
Re	α^*
100	29.453º
200	19.471º
300	15.215º

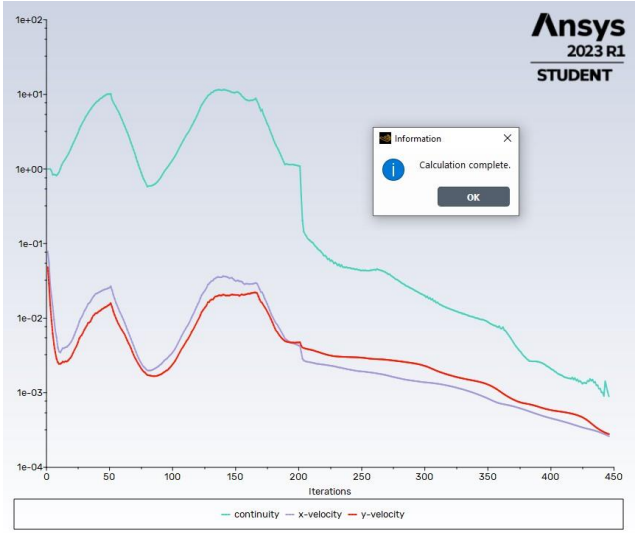


4.- ANSYS FLUENT. Definición del problema

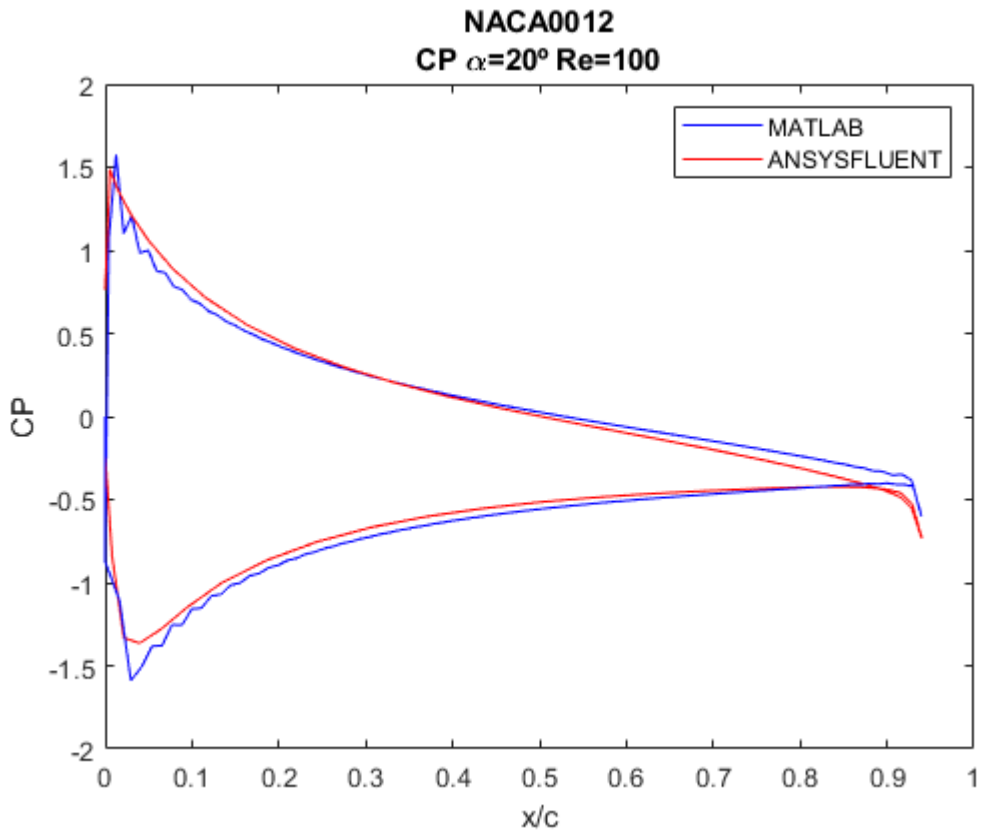


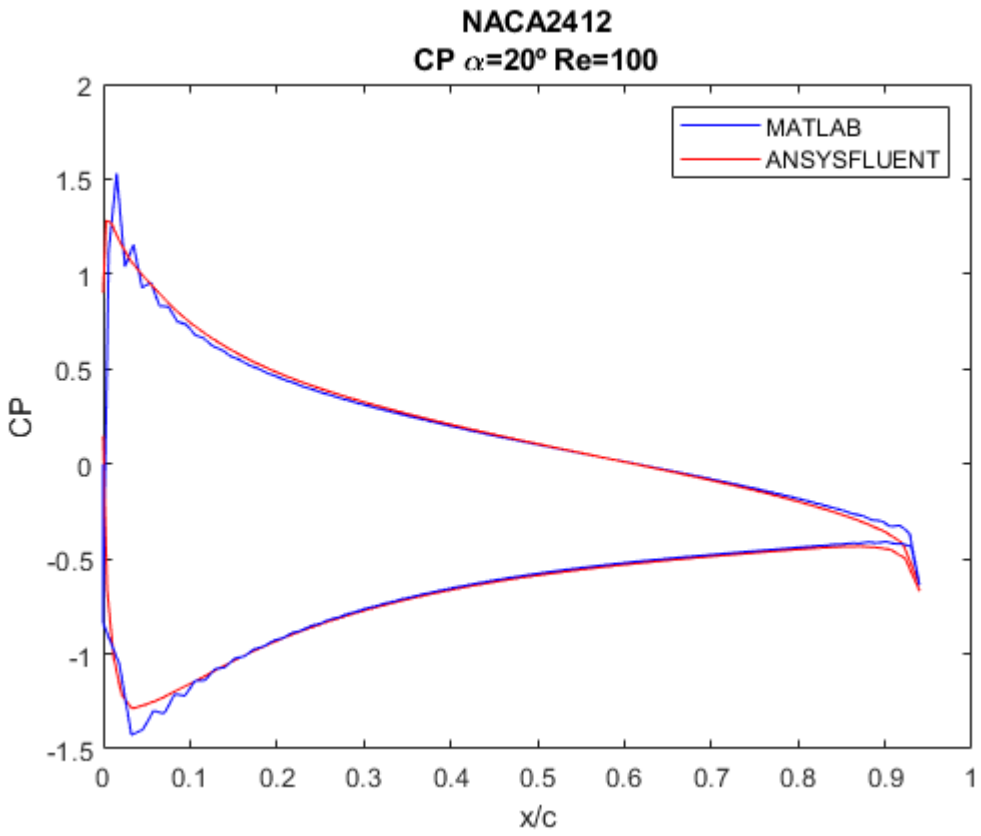






4.- ANSYS FLUENT. Comparación de resultados





4.- ANSYS FLUENT. Comparación de resultados

NACA 0012

C_D

Re	MATLAB	ANSYS FLUENT	**Error absoluto (ε_a)**	**Error relativo (ε_r)**
100	0.572197	0.585057	**0.01286**	**0.02198**
150	0.483168	0.490418	**0.00725**	**0.01478**
200	0.430491	0.434963	**0.004472**	**0.01028**

C_L

Re	MATLAB	ANSYS FLUENT	**Error absoluto (ε_a)**	**Error relativo (ε_r)**
100	0.740846	0.788417	**0.04757**	**0.06030**
150	0.684182	0.734051	**0.04987**	**0.06794**
200	0.636647	0.681264	**0.04462**	**0.06549**

NACA 2412

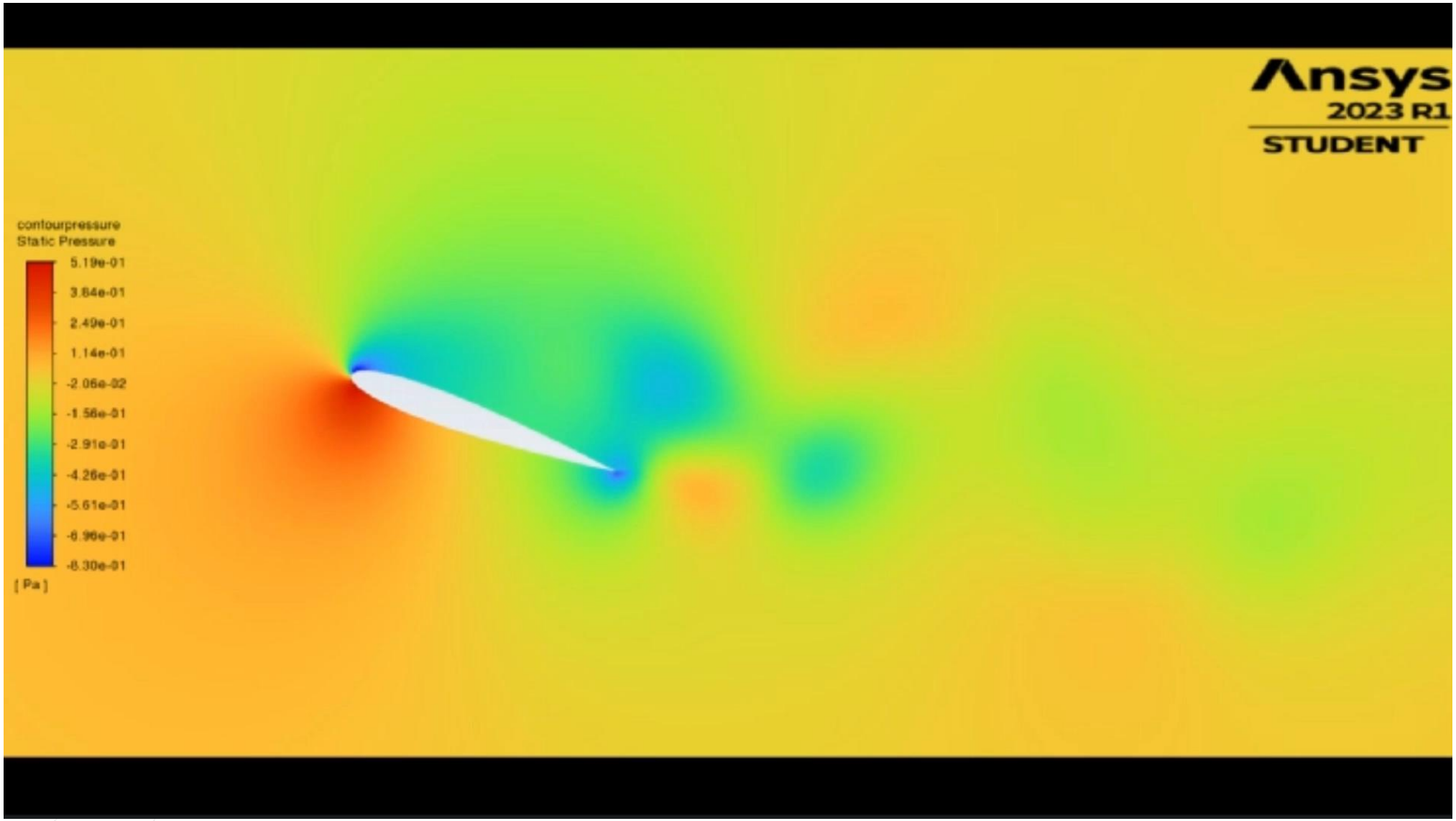
C_D

Re	MATLAB	ANSYS FLUENT	**Error absoluto (ε_a)**	**Error relativo (ε_r)**
100	0.572777	0.579380	**0.006603**	**0.001140**
150	0.484308	0.481448	**0.002860**	**0.005940**
200	0.432102	0.436183	**0.004080**	**0.009355**

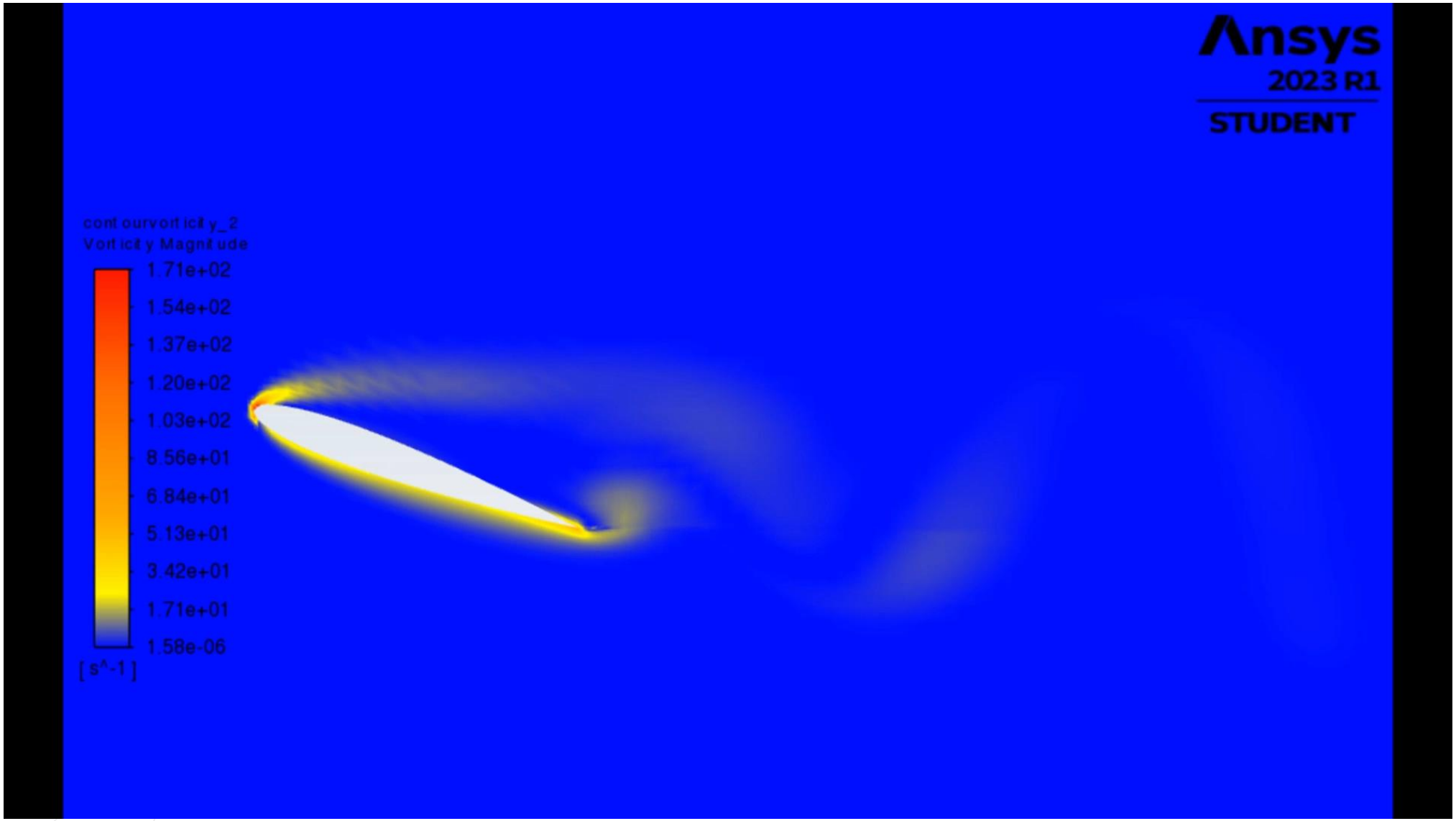
C_L

Re	MATLAB	ANSYS FLUENT	**Error absoluto (ε_a)**	**Error relativo (ε_r)**
100	0.797032	0.849898	**0.05287**	**0.06220**
150	0.749086	0.773213	**0.02413**	**0.03120**
200	0.707686	0.747337	**0.03965**	**0.05306**

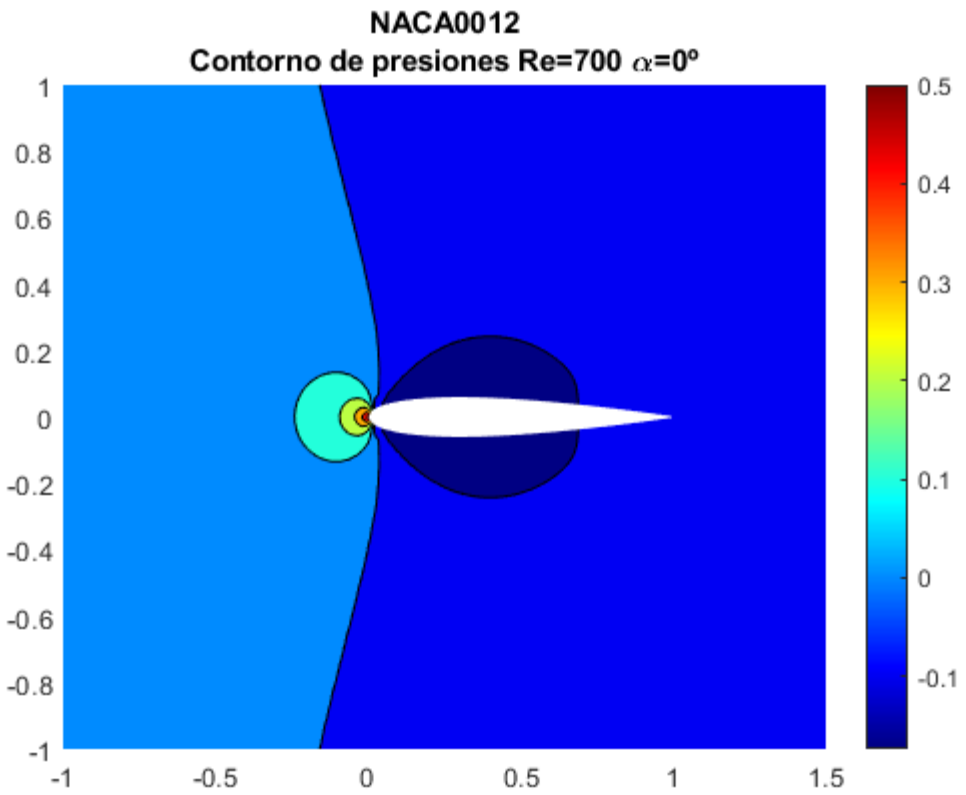
4.- ANSYS FLUENT. Régimen transitorio

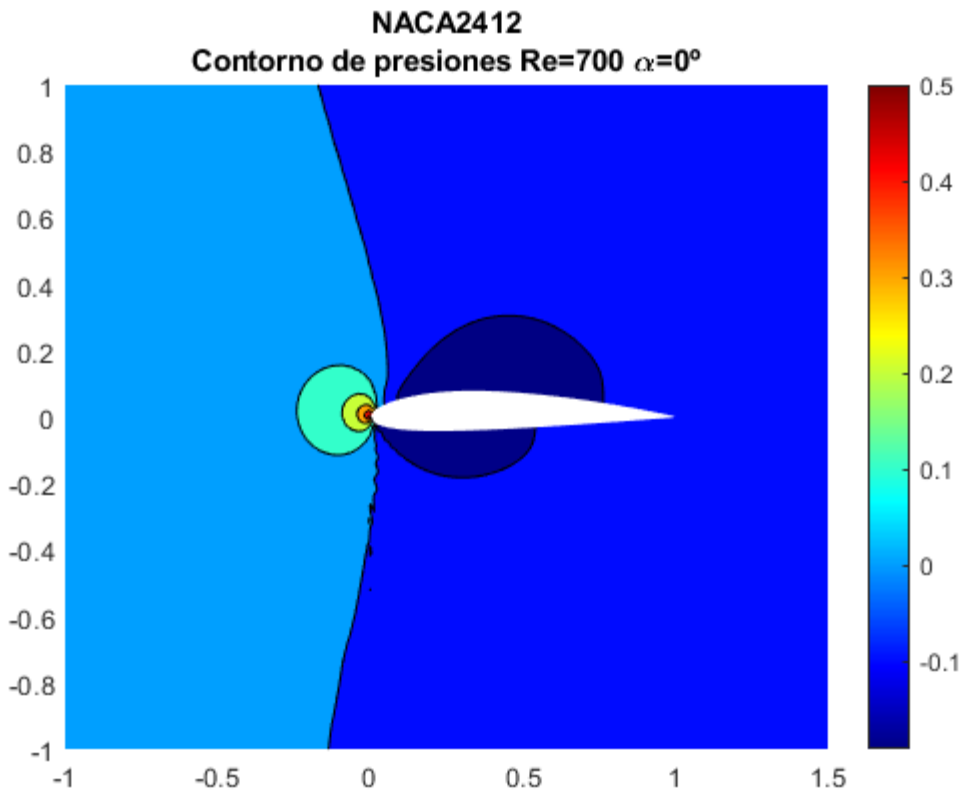


4.- ANSYS FLUENT. Régimen transitorio

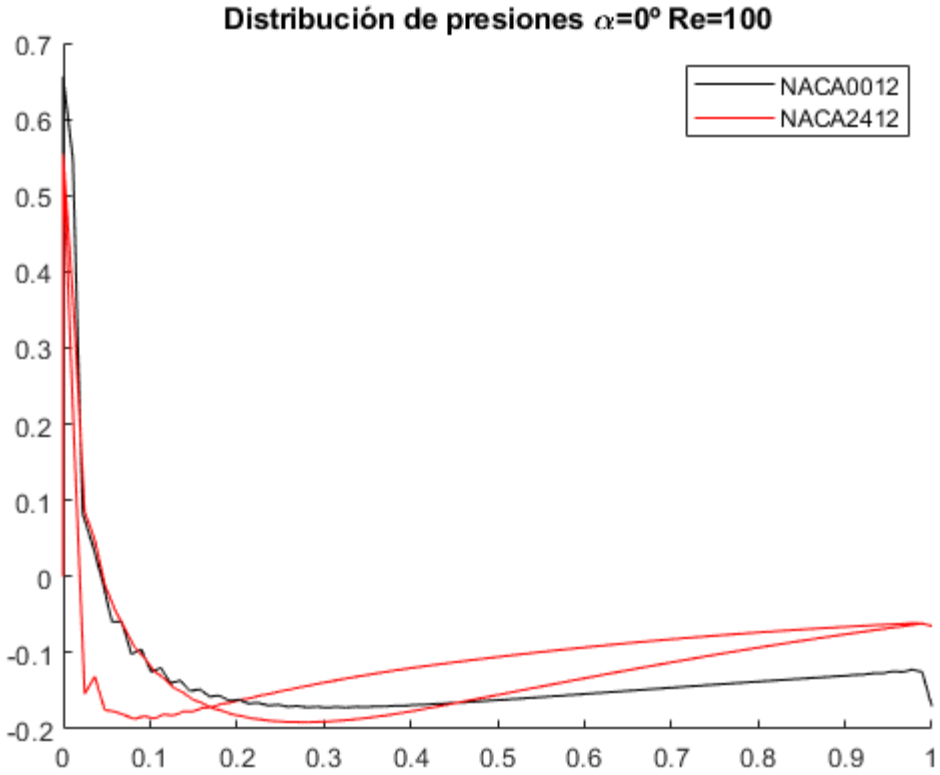


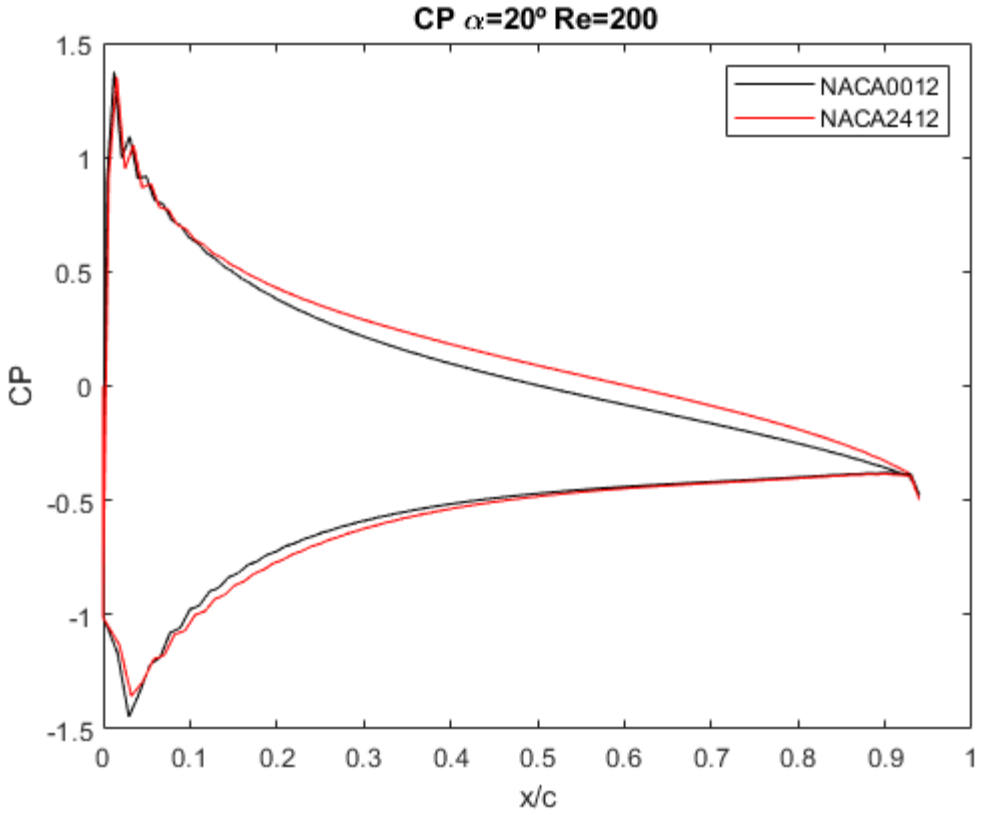
5.- Comparación. Flujo básico



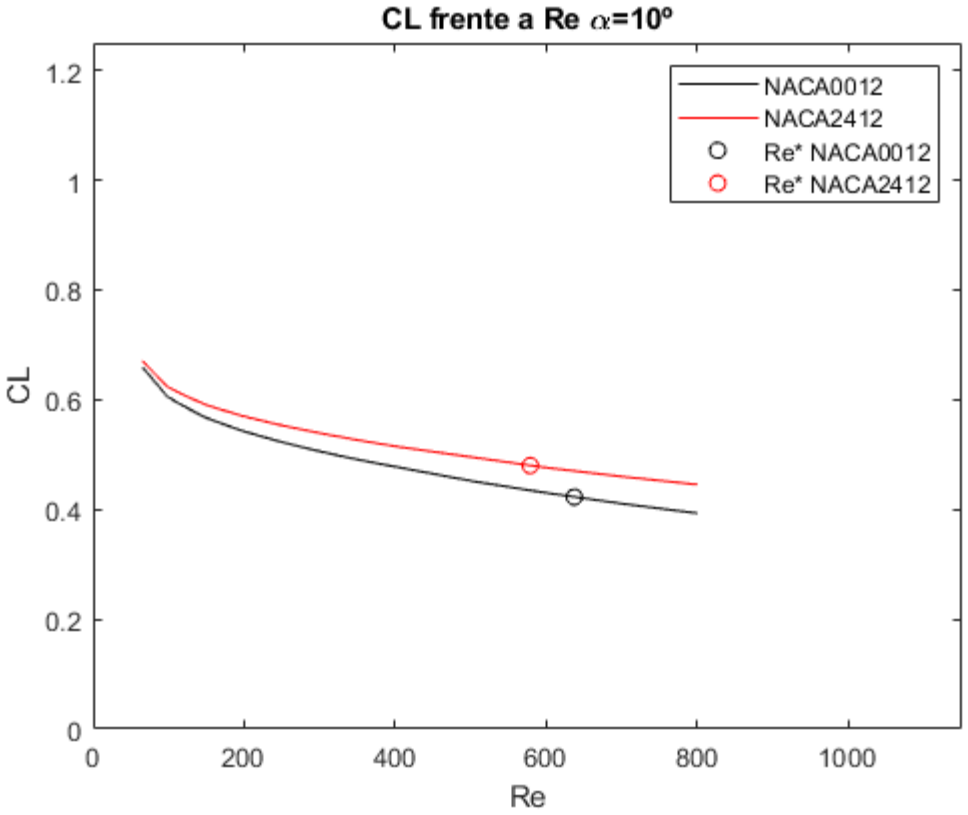


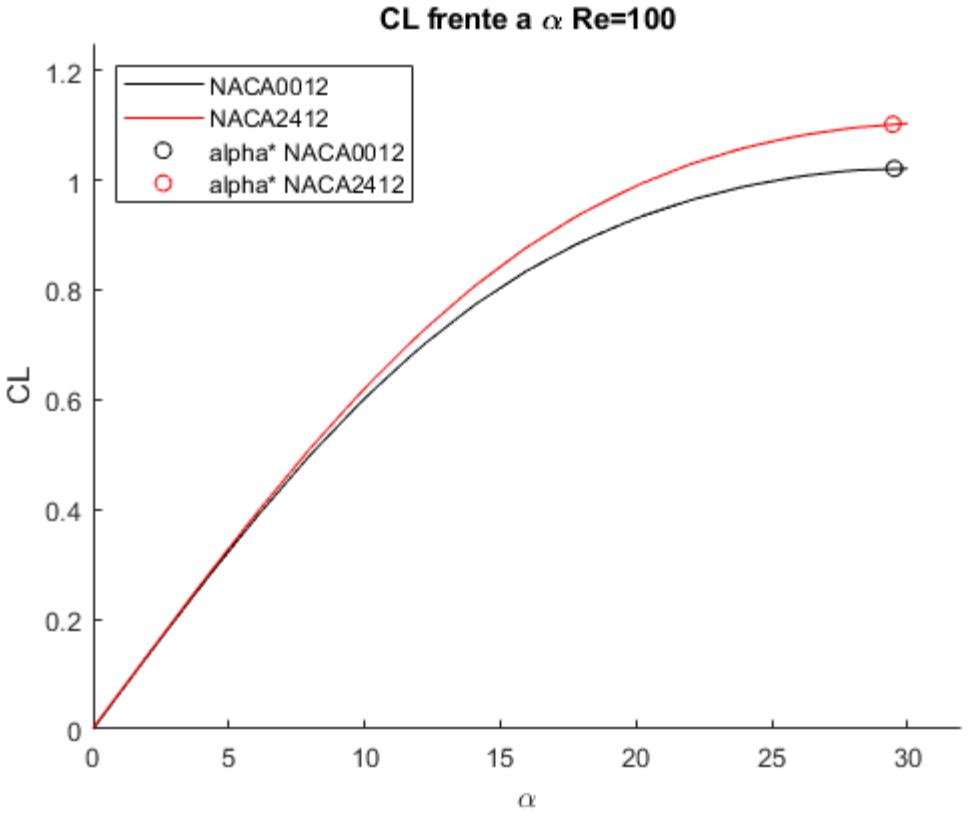
5.- Comparación. Flujo básico



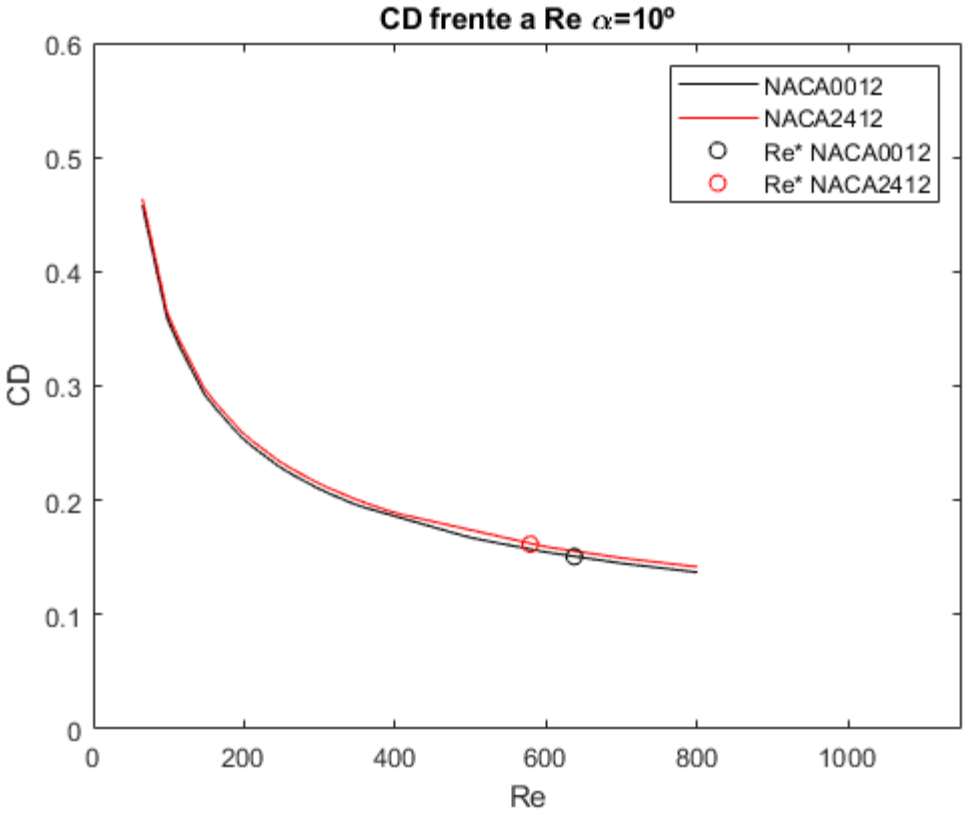


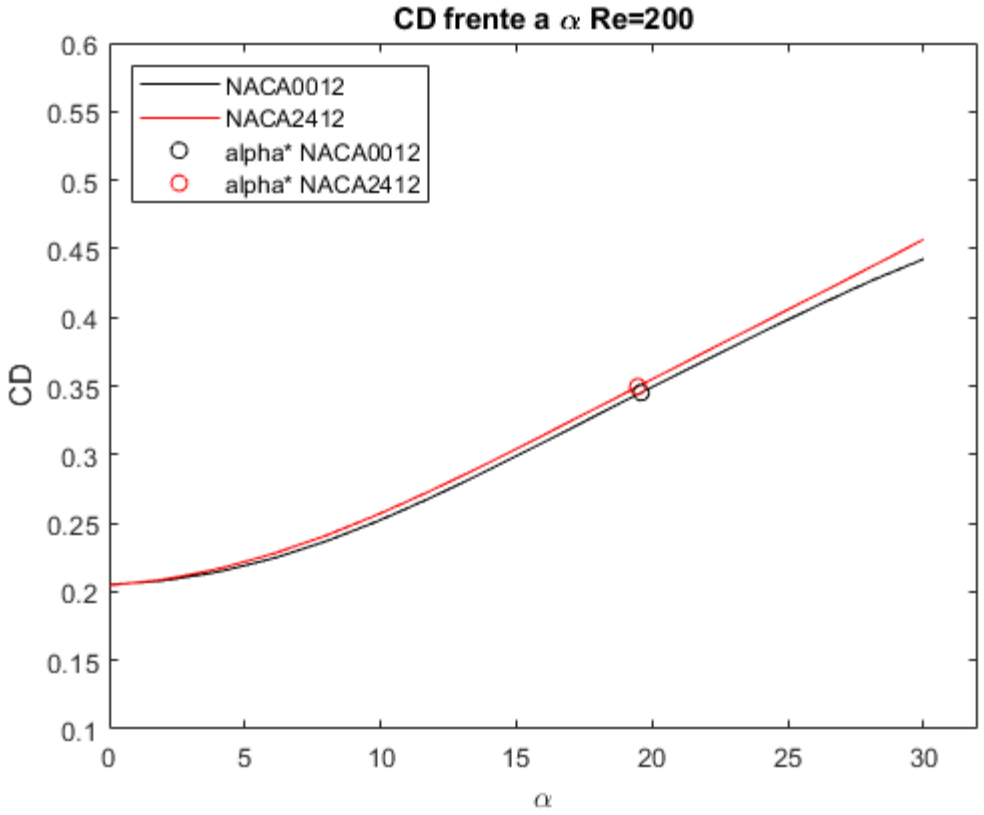
5.- Comparación. Flujo básico





5.- Comparación. Flujo básico

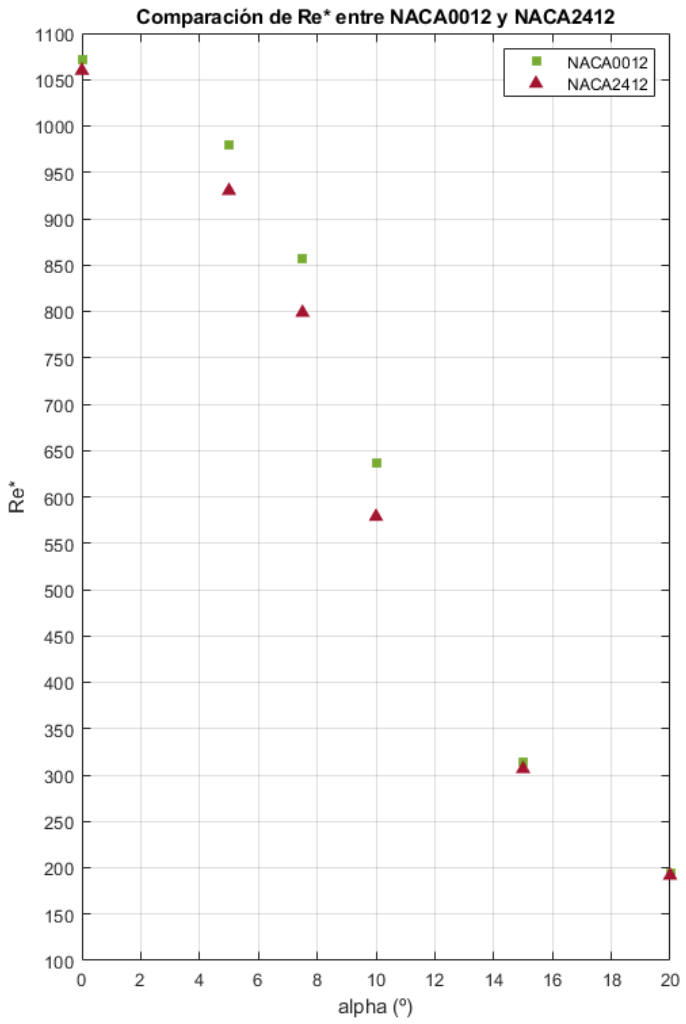


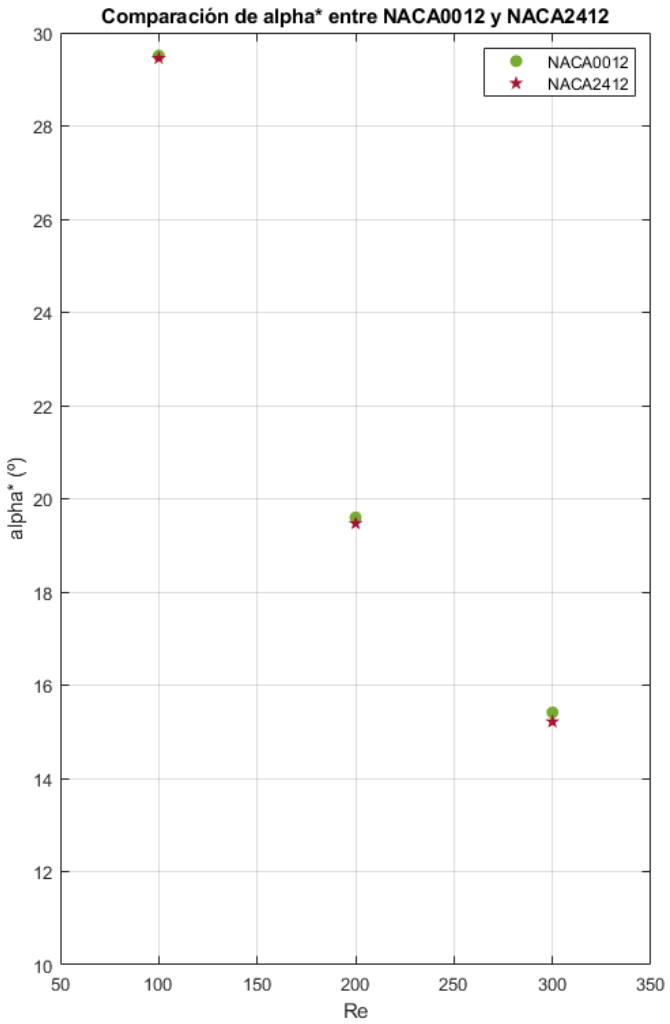


5.- Comparación. Estabilidad

α	Re^*	
	NACA0012	**NACA2412**
0º	1071.500	1059.780
5º	979.750	930.197
7.5º	856.211	798.090
10º	636.953	578.706
15º	313.895	306.594
20º	193.652	191.680

Re	α^*	
	NACA0012	**NACA2412**
100	29.509º	29.453º
200	19.598º	19.471
300	15.415º	15.215





6.- Conclusiones y líneas futuras

Conclusiones

- MATLAB VS ANSYS FLUENT
- NACA 0012 VS NACA 2412
  + Características
  + Aplicación
- Utilidad de los códigos desarrollados

Líneas futuras

- Obtención de un mejor mallado
- Comprobación estabilidad con régimen transitorio

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2.- Formulación del problema. Optimización

- Según el número de puntos horizontales

MALLA	80-50-50	80-71-85	120-71-85	150-71-85	120-50-85
c_{dp}	0.239	0.2449	0.2491	0.2507	0.2434
c_{dv}	0.1207	0.1119	0.1026	0.1000	0.1106
c_{lp}	0.6422	0.6526	0.6517	0.6507	0.6439
c_{lv}	0.1285	0.1199	0.1108	0.1080	0.1203

MALLA	120-90-85	120-71-50	120-71-120	150-90-120	**150-200-120**
c_{dp}	0.2515	0.2483	0.2493	0.2536	**0.2543**
c_{dv}	0.0984	0.1032	0.1024	0.0957	**0.0839**
c_{lp}	0.6551	0.6494	0.6522	0.6549	**0.6596**
c_{lv}	0.1066	0.1105	0.1109	0.1039	**0.0948**

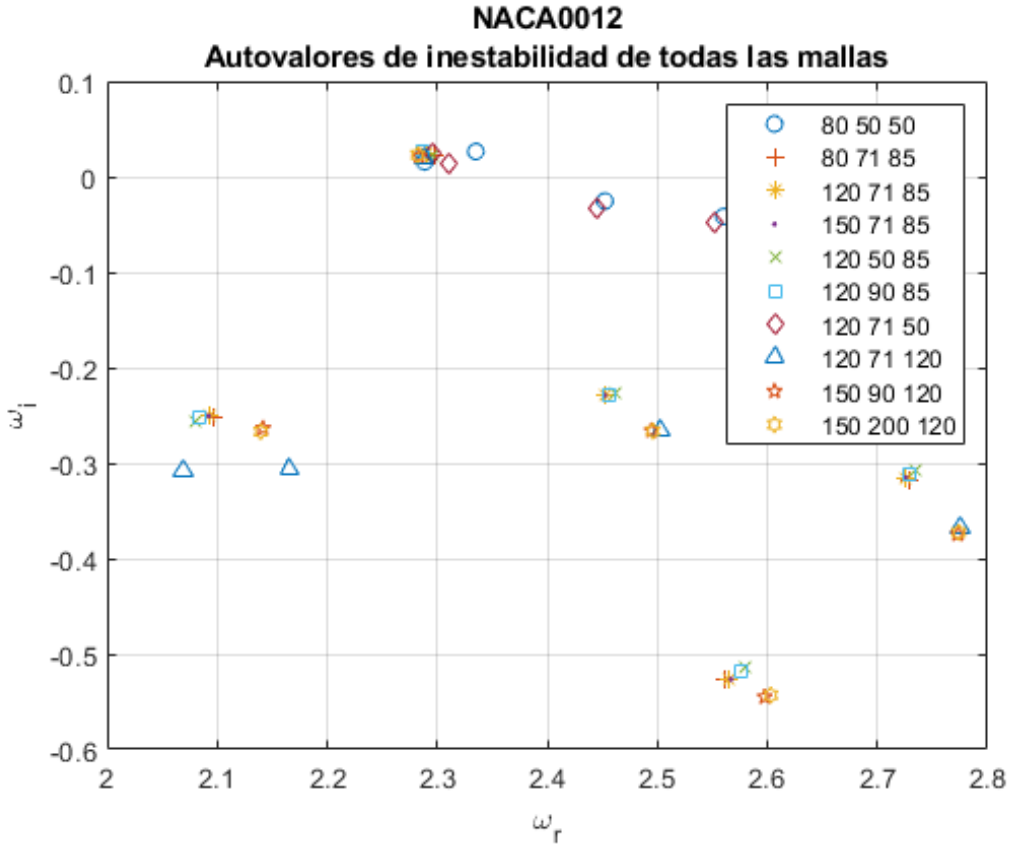
MALLA	80-50-50	80-71-85	120-71-85	**150-71-85**	120-50-85
$\varepsilon_{c_{dp}}$	0.0602	0.0370	0.0204	**0.0142**	0.0429
$\varepsilon_{c_{dv}}$	0.4386	0.3337	0.2229	**0.1919**	0.3182
$\varepsilon_{c_{lp}}$	0.0264	0.0106	0.0120	**0.0135**	0.0238
$\varepsilon_{c_{lv}}$	0.3555	0.2648	0.1688	**0.1392**	0.2690

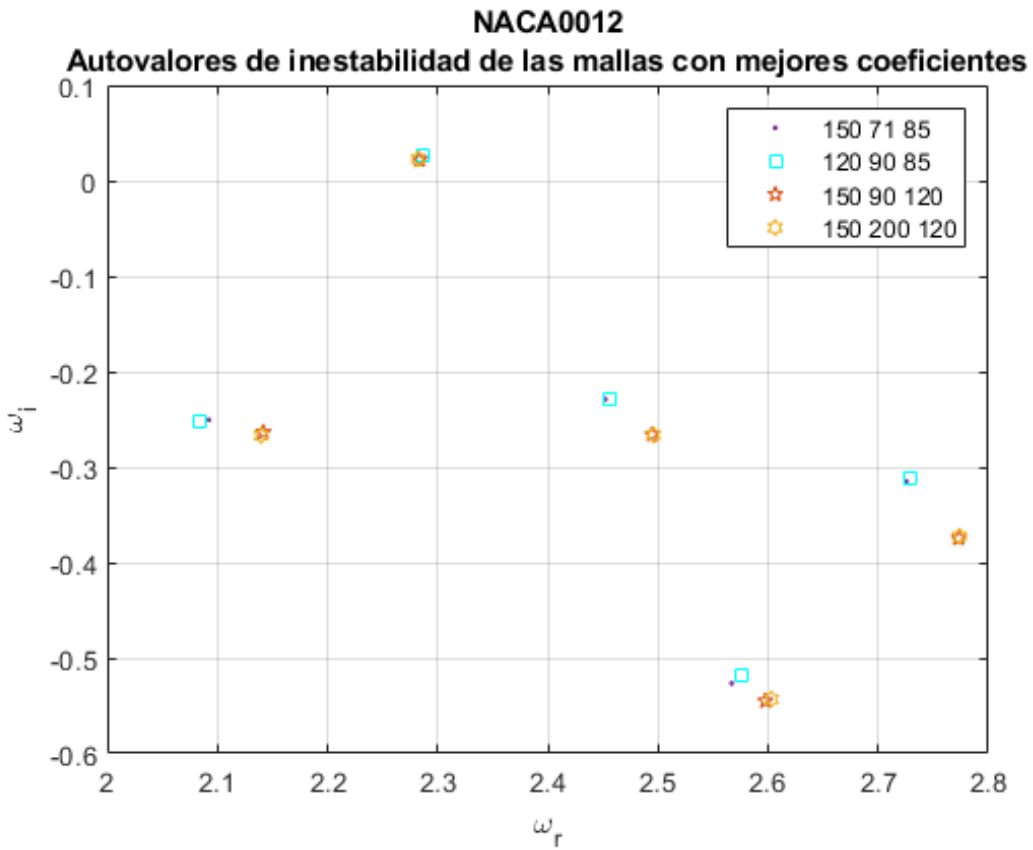
MALLA	**120-90-85**	120-71-50	120-71-120	**150-90-120**
$\varepsilon_{c_{dp}}$	**0.0110**	0.0236	0.0197	**0.0028**
$\varepsilon_{c_{dv}}$	**0.1728**	0.2300	0.2205	**0.1406**
$\varepsilon_{c_{lp}}$	**0.0068**	0.0155	0.0112	**0.0071**
$\varepsilon_{c_{lv}}$	**0.1245**	0.1656	0.1698	**0.0960**

MALLA	1ª Iteración	2ª Iteración	3ª Iteración	**Media**
150-71-85	211.61 s	210.38 s	211.52 s	**211.17 s**
120-90-85	186.36 s	182.90 s	184.90 s	**184.72 s**
150-90-120	297.01 s	298.70 s	297.60 s	**297.77 s**
150-200-120	454.66 s	464.17 s	463.38 s	**460.74 s**

2.- Formulación del problema. Optimización

- Según el número de puntos horizontales





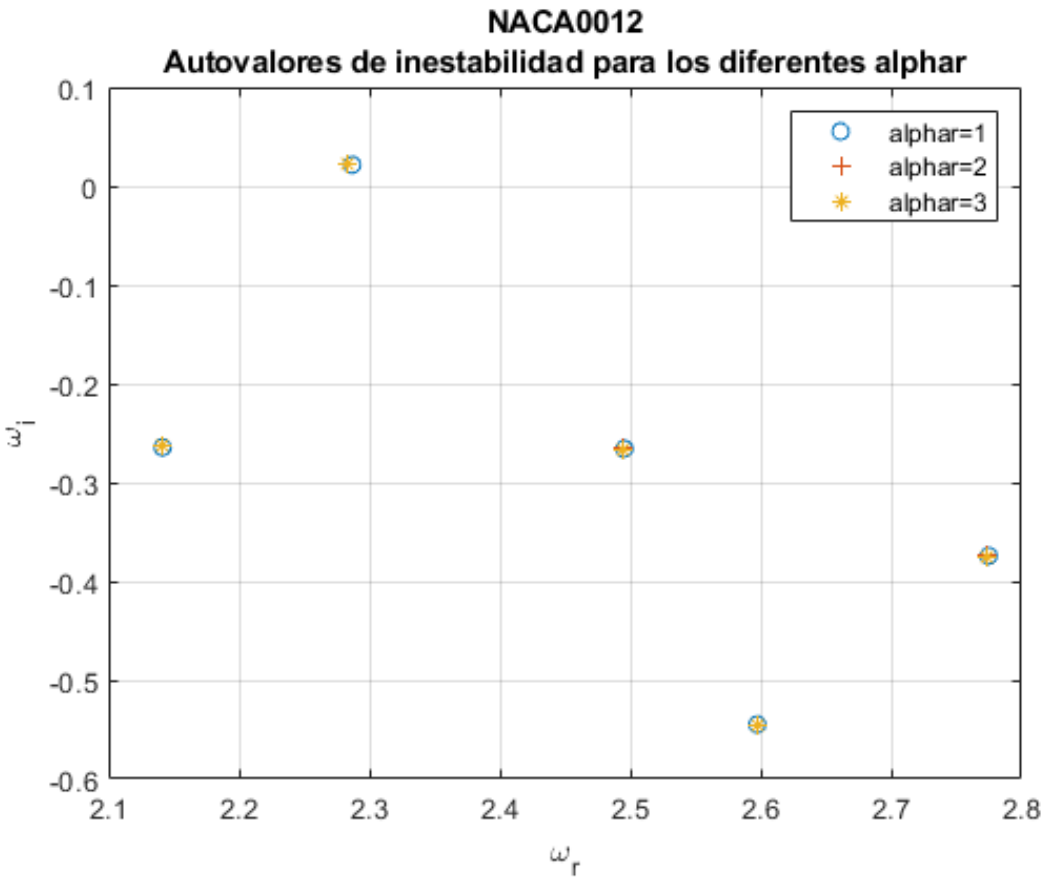
MALLA	Primero	Segundo	Tercero	Cuarto	Quinto	**Máx. error por malla**
150-71-85	0.05008	0.00602	0.05783	0.03972	0.07500	**0.07500**
120-90-85	0.05805	0.00609	0.05480	0.03560	0.07635	**0.07635**
150-90-120	0.00395	0.00174	0.00137	0.00590	0.00205	**0.00590**

2.- Formulación del problema. Optimización

- Según el concentrador de puntos

alphar	1	2	**3**
c_{dp}	0.2516	0.2536	**0.2544**
c_{dv}	0.0998	0.0957	**0.0949**
c_{lp}	0.6551	0.6549	**0.6550**
c_{lv}	0.1069	0.1039	**0.1024**

alphar	1	2
$\varepsilon_{c_{dp}}$	0.00983	0.00314
$\varepsilon_{c_{dv}}$	0.05163	0.00843
$\varepsilon_{c_{lp}}$	0.00015	0.00015
$\varepsilon_{c_{lv}}$	0.04395	0.01465



alphar	1ª Iteración	2ª Iteración	3ª Iteración	**Media**
1	290.75 s	289.90 s	288.44 s	**289.70 s**
2	297.01 s	298.70 s	297.60 s	**297.77 s**
3	330.47 s	337.74 s	336.27 s	**334.83 s**

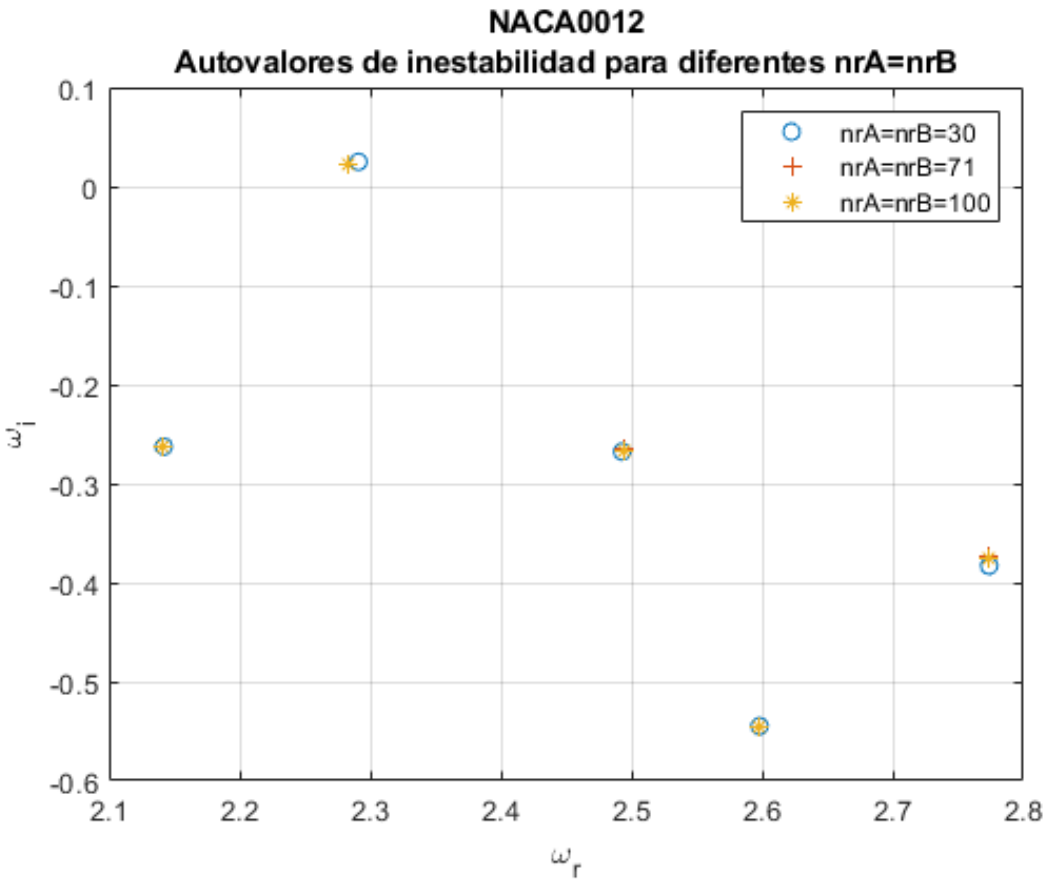
alphar	Primero	Segundo	Tercero	Cuarto	Quinto	**Máx. error por malla**
1	0.00094	0.00424	0.00110	0.00020	0.002114	**0.00424**
2	0.00021	0.00091	0.00022	0.00004	0.00044	**0.00091**

2.- Formulación del problema. Optimización

- Según el número de puntos verticales

$n_{rA}=n_{rB}$	30	71	**100**
c_{dp}	0.2522	0.2536	**0.2541**
c_{dv}	0.1020	0.0957	**0.0952**
c_{lp}	0.6569	0.6549	**0.6549**
c_{lv}	0.1083	0.1039	**0.1031**

$n_{rA}=n_{rB}$	30	71
$\varepsilon_{c_{dp}}$	0.00748	0.00197
$\varepsilon_{c_{dv}}$	0.07143	0.00525
$\varepsilon_{c_{lp}}$	0.00305	0
$\varepsilon_{c_{lv}}$	0.05044	0.00776



$n_{rA}=n_{rB}$	1ª Iteración	2ª Iteración	3ª Iteración	**Media**
30	60.49 s	59.91 s	61.44 s	**61.61 s**
71	297.01 s	298.70 s	297.60 s	**297.77 s**
100	896.29 s	966.22 s	912.73 s	**925.08 s**

alphar	Primero	Segundo	Tercero	Cuarto	Quinto	**Máx. error por malla**
1	0.00094	0.00424	0.00110	0.00020	0.002114	**0.00424**
2	0.00021	0.00091	0.00022	0.00004	0.00044	**0.00091**

4.- ANSYS FLUENT. Definición perfil (Ecuaciones)

	Front ($0 \le x < p$)	Back ($p \le x \le 1$)
Camber	$y_c = \frac{M}{P^2}\left(2Px - x^2\right)$	$y_c = \frac{M}{(1-P)^2}\left(1 - 2P + 2Px - x^2\right)$
Gradient	$\frac{dy_c}{dx} = \frac{2M}{P^2}(P - x)$	$\frac{dy_c}{dx} = \frac{2M}{(1-P)^2}(P - x)$

$$\theta = \operatorname{atan}\left(\frac{dy_c}{dx}\right)$$

Upper Surface	$x_u = x_c - y_t \sin(\theta)$	$y_u = y_c + y_t \cos(\theta)$
Lower Surface	$x_l = x_c + y_t \sin(\theta)$	$y_l = y_c - y_t \cos(\theta)$

The thickness distribution is given by the equation:

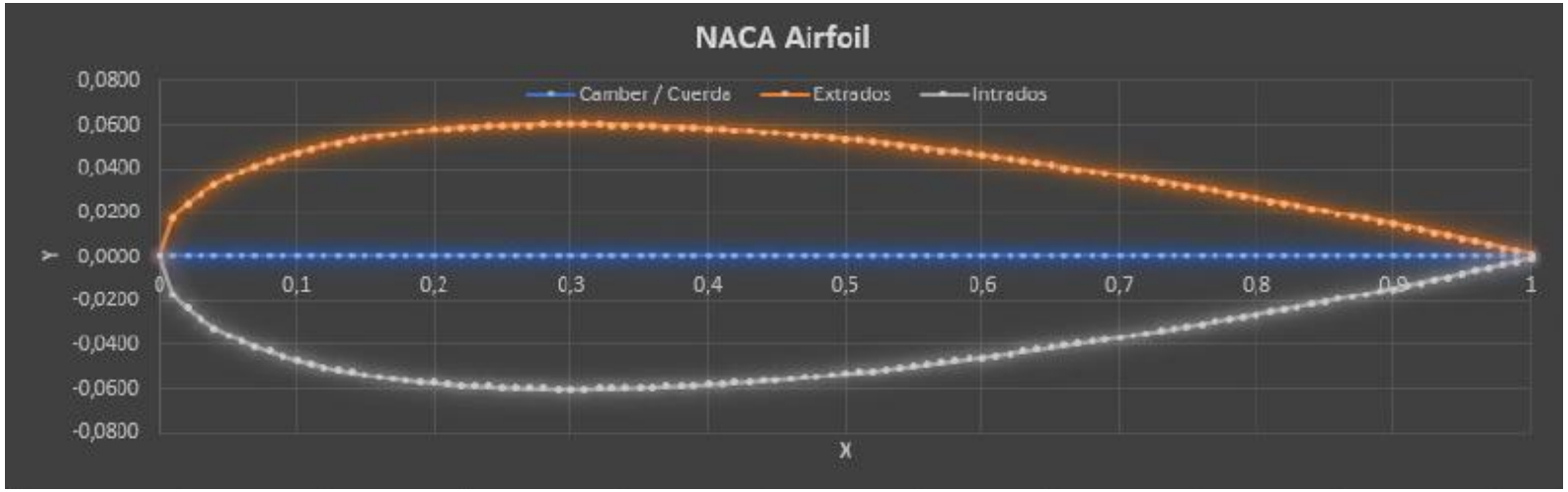
$$y_t = \frac{T}{0.2}\left(a_0 x^{0.5} + a_1 x + a_2 x^2 + a_3 x^3 + a_4 x^4\right)$$

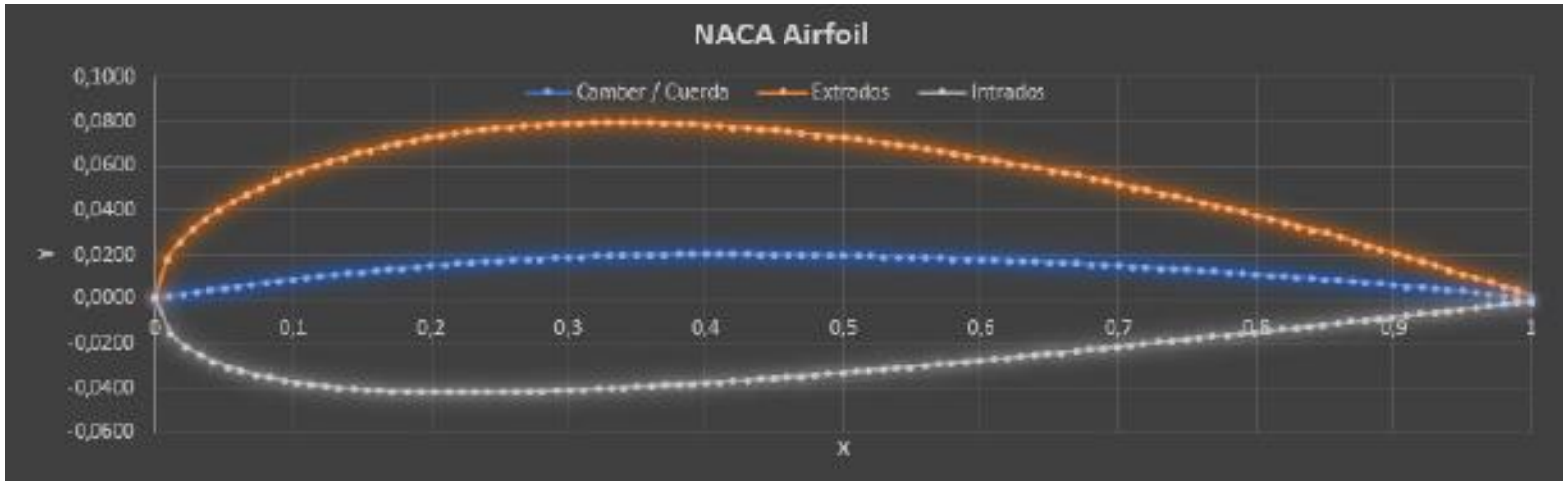
Where:

$a_0 = 0.2969$ $a_1 = -0.126$ $a_2 = -0.3516$ $a_3 = 0.2843$

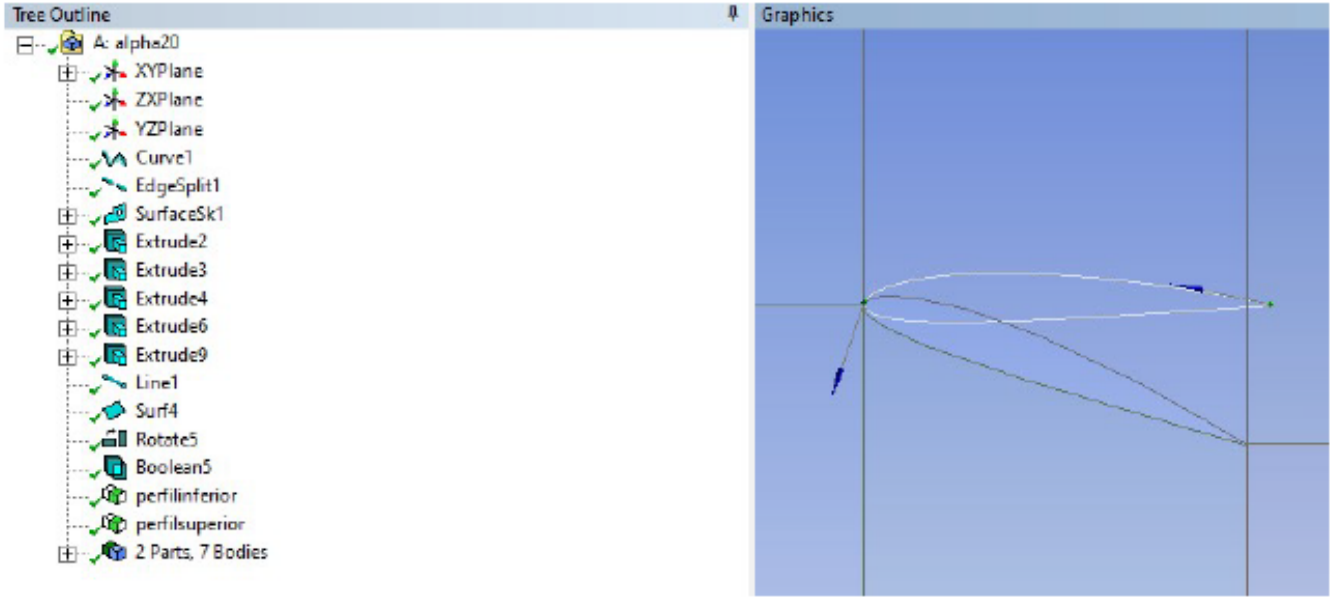
$a_4 = -0.1015$ or -0.1036 for a closed trailing edge

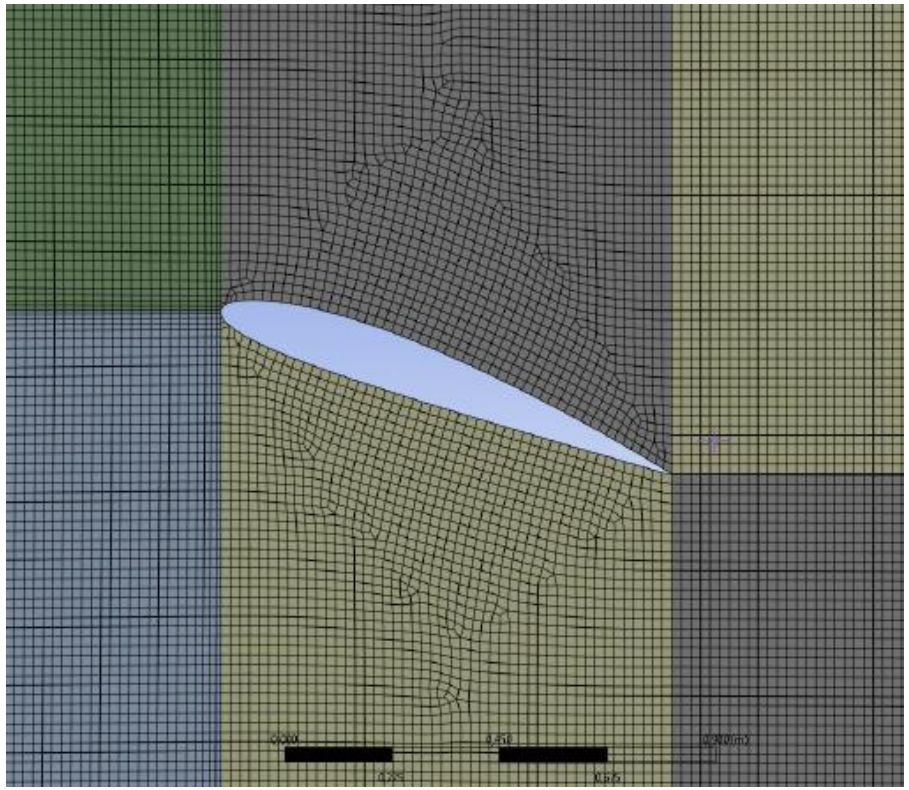
4.- ANSYS FLUENT. Definición perfil (Excel)



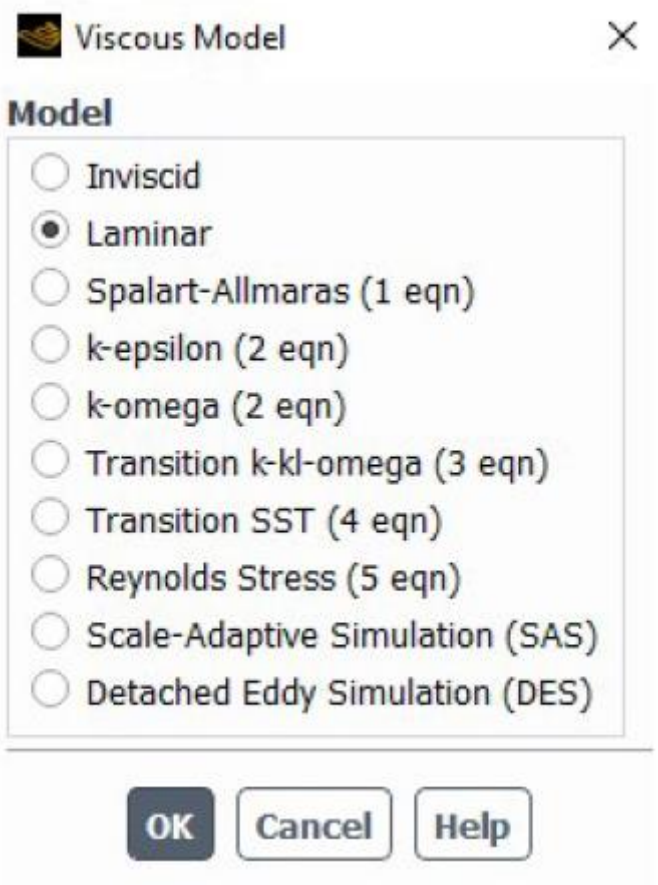


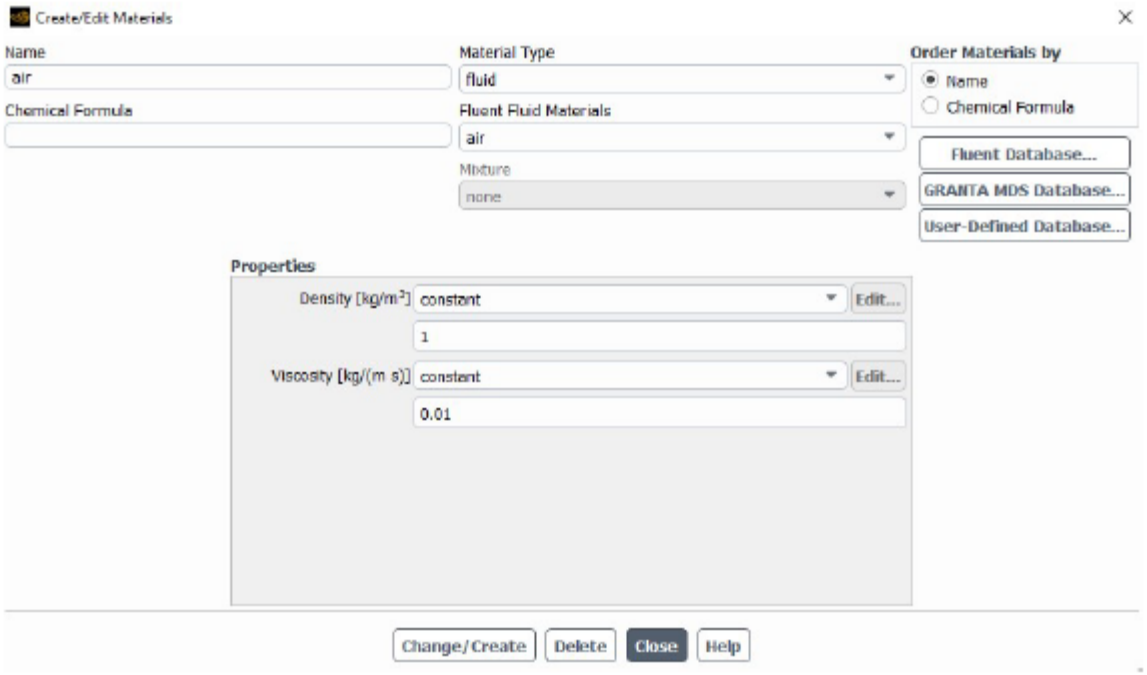
4.- ANSYS FLUENT. Definición perfil NACA2412





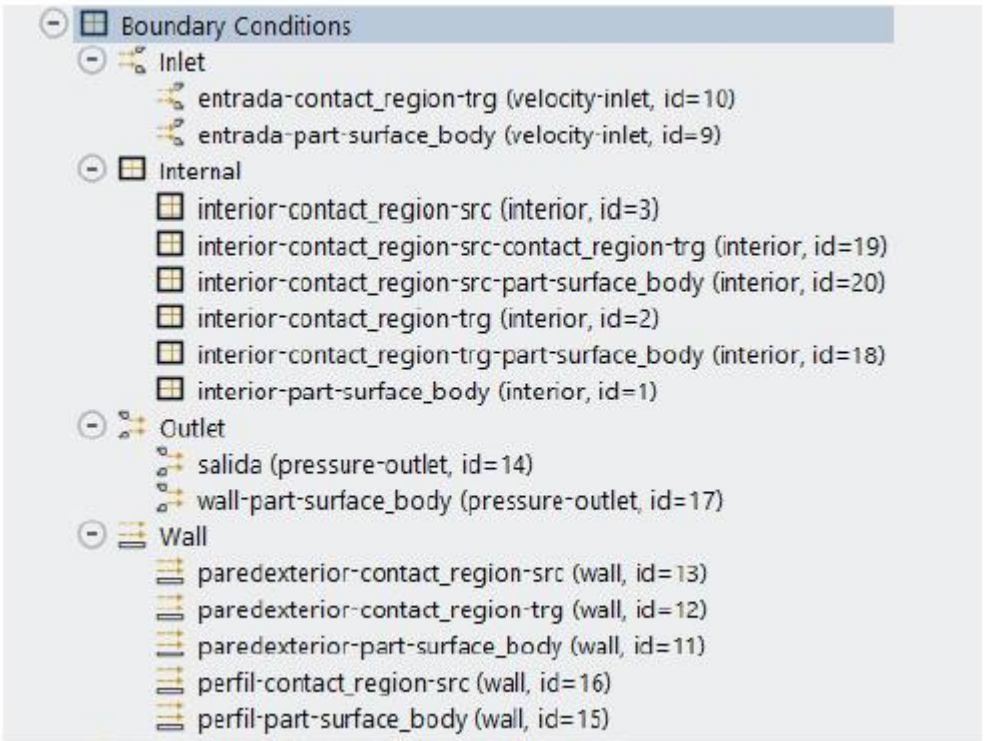
4.- ANSYS FLUENT. Definición del fluido

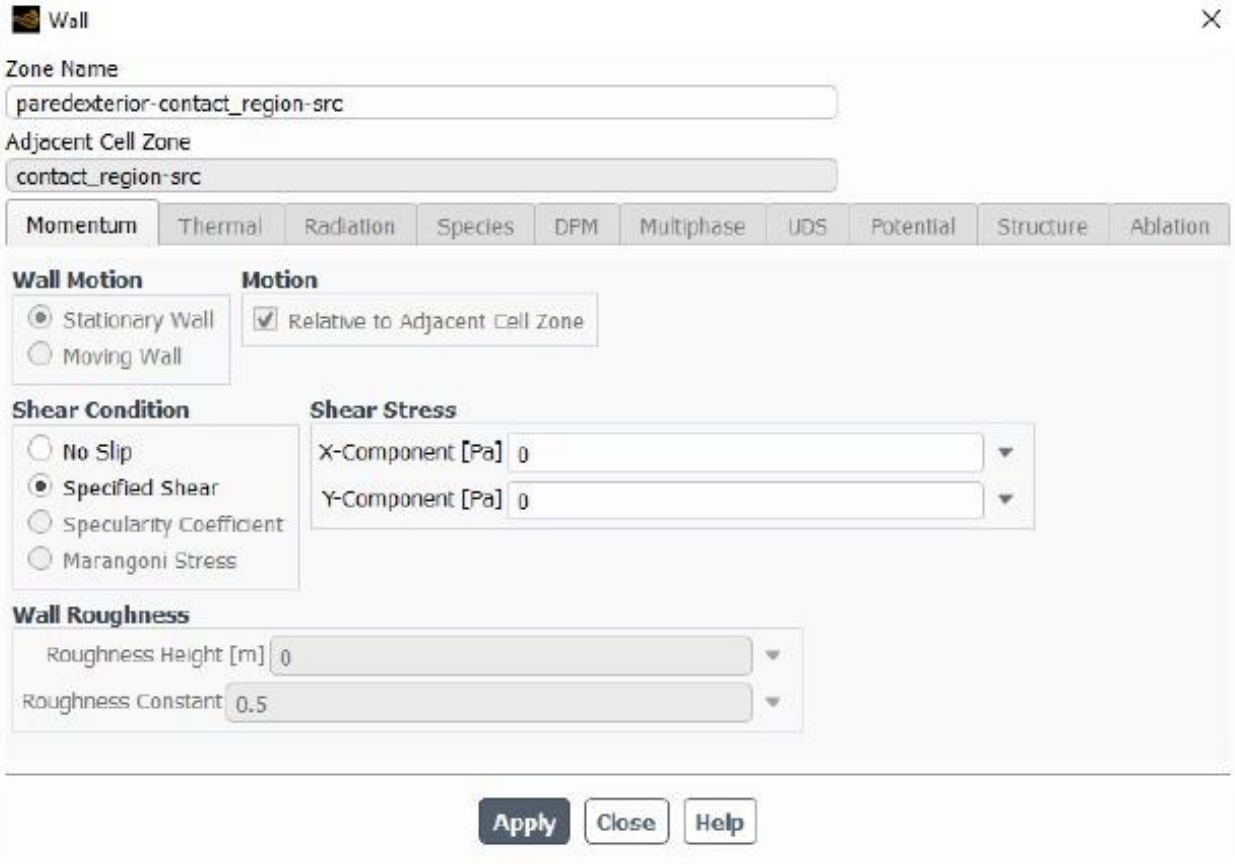




$$Re = \frac{\rho \cdot U_\infty \cdot L}{\mu}$$

4.- ANSYS FLUENT. Condiciones de contorno





4.- ANSYS FLUENT. Métodos numéricos

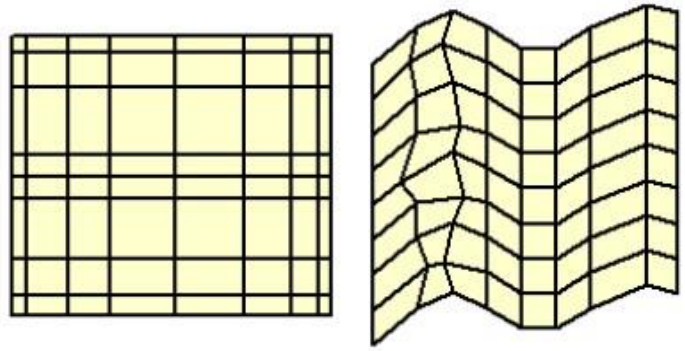
Discretización espacial (Volúmenes finitos)

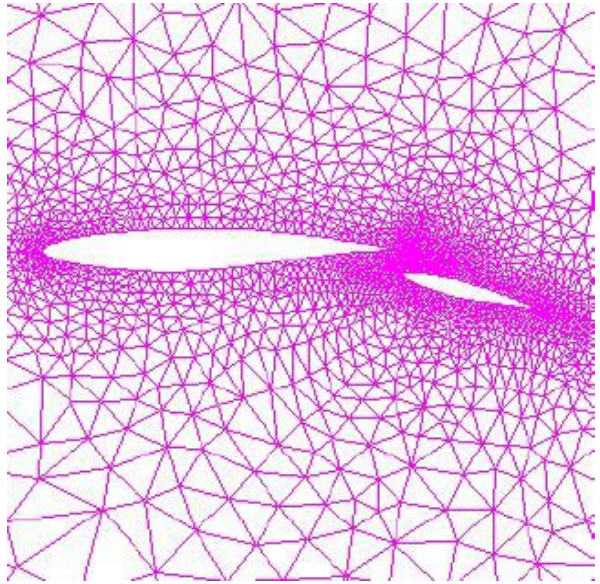
Ventajas:

- Fáciles de implementar
- Bajo coste computacional
- Si el mallado es estructurado: matrices dispersas con estructura en banda
- Muy flexibles con la geometría

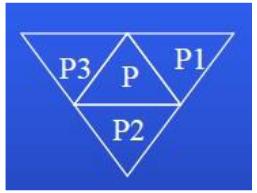
Inconvenientes:

- Difícil implementación más allá del tercer orden espacial

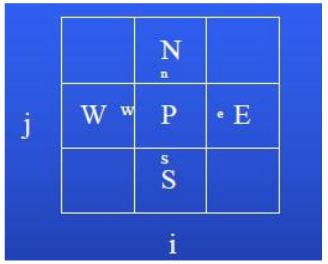




4.- ANSYS FLUENT. Métodos numéricos



$$\int_{\Sigma_1+\Sigma_2+\Sigma_3} \Phi\vec{v}\cdot\vec{n}\,d\sigma \approx \Phi_1(\vec{v}\cdot\vec{n})_1 d\sum_1 + \Phi_2(\vec{v}\cdot\vec{n})_2 d\sum_2 + \Phi_3(\vec{v}\cdot\vec{n})_3 d\sum_3$$



$$\int_{\Sigma_e+\Sigma_w+\Sigma_s+\Sigma_n} \Phi\vec{v}\cdot\vec{n}\,d\sigma \approx \Phi_e(\vec{v}\cdot\vec{n})_e d\sum_e + \Phi_w(\vec{v}\cdot\vec{n})_w d\sum_w + \Phi_s(\vec{v}\cdot\vec{n})_s d\sum_s + \Phi_n(\vec{v}\cdot\vec{n})_n d\sum_n$$

Técnicas de interpolación (Volúmenes finitos)

$$F_e = \int_{\Sigma_e} \Phi\vec{v}\cdot\vec{n}\,d\sigma \approx \Phi_e(\vec{v}\cdot\vec{n})_e d\sum_e = \Phi_e v_e d\sum_e \rightarrow \Phi_e ?$$

- Upwind

$$\Phi_e = \begin{cases} \Phi_P & \text{si } v_e > 0 \\ \Phi_E & \text{si } v_e < 0 \end{cases}$$

- Lineal

$$\Phi_e = \Phi_E \lambda_e + \Phi_P(1-\lambda_e)$$

$$\lambda_e = \frac{x_e - x_P}{x_E - x_P}$$

- Cuadrática (QUICK)

$$\Phi_e = \begin{cases} f(\Phi_E,\Phi_P,\Phi_W) & \text{si } v_e > 0 \\ f(\Phi_E,\Phi_P,\Phi_E) & \text{si } v_e < 0 \end{cases}$$

- Deferred Correction (en métodos iterativos)

$$F_e = F_e^L + (F_e^H - F_e^L)^{old}$$

4.- ANSYS FLUENT. Métodos numéricos

Factores de relajación

$$\Phi^{(k+1)} = \Phi^k + \alpha(\Phi^* - \Phi^k)$$

Residuos y convergencia

- Una vez linealizada las ecuaciones:

$$a_e \Phi_e = \sum_{nb} a_{ne} \Phi_{ne} + b$$

- Residuo escalado:

$$R^{\Phi} = \sum_{celdas} \left| \sum_{nb} a_{ne} \Phi_{ne} + b - a_{ne} \Phi_e \right| / \sum_{celdas} a_e \Phi_e$$